Answer Key

Other (142 Questions)

Q1. (B)	Q2. (B)	Q3. (B)	Q4. (C)	Q5. (C)
Q6. (A)	Q7. (A)	Q8. (D)	Q9. (B)	Q10. (C)
Q11. (B)	Q12. (B)	Q13. (C)	Q14. (A)	Q15. (B)
Q16. (D)	Q17. (A)	Q18. (C)	Q19. (A)	Q20. (A)
Q21. (A)	Q22. (C)	Q23. (A)	Q24. (D)	Q25. (A)
Q26. (C)	Q27. (C)	Q28. (C)	Q29. (B)	Q30. (C)
Q31. (D)	Q32. (B)	Q33. (A)	Q34. (D)	Q35. (A)
Q36. (B)	Q37. (B)	Q38. (D)	Q39. (C)	Q40. (A)
Q41. (B)	Q42. (A)	Q43. (A)	Q44. (C)	Q45. (D)
Q46. (A)	Q47. (A)	Q48. (A)	Q49. (D)	Q50. (D)
Q51. (C)	Q52. (C)	Q53. (B)	Q54. (D)	Q55. (A)
Q56. (C)	Q57. (C)	Q58. (D)	Q59. (C)	Q60. (B)
Q61. (B)	Q62. (D)	Q63. (A)	Q64. (D)	Q65. (B)
Q66. (A)	Q67. (D)	Q68. (B)	Q69. (A)	Q70. (A)
Q71. (C)	Q72. (A)	Q73. (B)	Q74. (C)	Q75. (B)
Q76. (B)	Q77. (D)	Q78. (A)	Q79. (B)	Q80. (B)
Q81. (D)	Q82. (D)	Q83. (B)	Q84. (B)	Q85. (A)
Q86. (C)	Q87. (B)	Q88. (A)	Q89. (D)	Q90. (A)
Q91. (B)	Q92. (B)	Q93. (C)	Q94. (D)	Q95. (A)
Q96. (A)	Q97. (A)	Q98. (C)	Q99. (D)	Q100.(A)
Q101.(D)	Q102.(B)	Q103.(C)	Q104.(B)	Q105.(B)

Q106.(B)	**Q107.(D)**	**Q108.(D)**	**Q109.(C)**	**Q110.(D)**
Q111.(B)	**Q112.(C)**	**Q113.(A)**	**Q114.(C)**	**Q115.(B)**
Q116.(A)	**Q117.(B)**	**Q118.(C)**	**Q119.(B)**	**Q120.(D)**
Q121.(D)	**Q122.(A)**	**Q123.(C)**	**Q124.(C)**	**Q125.(C)**
Q126.(C)	**Q127.(A)**	**Q128.(B)**	**Q129.(A)**	**Q130.(B)**
Q131.(C)	**Q132.(A)**	**Q133.(A)**	**Q134.(C)**	**Q135.(B)**
Q136.(D)	**Q137.(A)**	**Q138.(B)**	**Q139.(B)**	**Q140.(A)**
Q141.(B)	**Q142.(B)**			

Solutions

Q1. Solution

Correct Answer: (B)

Increase in kinetic energy = Decrease in potential energy

$$\therefore \quad \frac{1}{2}mv^2 = \frac{mgR}{1+\frac{R}{R}} = \frac{mgR}{2} \quad \left(\Delta U = \frac{mgh}{1+\frac{h}{R}}, h = R\right)$$

$$\Rightarrow \quad mv^2 = mgR \Rightarrow v = \sqrt{gR}$$

Q2. Solution

Correct Answer: (B)

From $i-t$ graph, area from $t=0$ to $t=2$ s, we get

$$= \frac{1}{2} \times 1 \times 10 + \frac{1}{2} \times (2-1) \times 10$$

$$= 5 + 5 = 10 \text{ A}$$

$$\therefore \text{ Area current, } i_{\text{av}} = \frac{\text{Area } (i-t) \text{ graph}}{\text{time interval}} = \frac{10}{2} = 5 \text{ A}$$

Q3. Solution

Correct Answer: (B)

Maximum acceleration of 1 kg block may be $a_{\max} = \mu g = 1 \text{ m/s}^2$ Common acceleration without relative motion between two blocks may be, $a = \frac{0.5}{3} \text{ m/s}^2$ Since, $a < a_{\max}$ There will be no relative motion and blocks will move with acceleration $\frac{0.5}{3} \text{ m/s}^2$. Force of friction by lower block on upper block,

$$f = ma = (1)\left(\frac{0.5}{3}\right) = \frac{1}{6} \text{ N(towards right)}$$

$$\therefore \quad W = f \times s$$

$$= \frac{1}{6} \times 3 = 0.5 \text{ J}$$

Q4. Solution

Correct Answer: (C)

$$\Delta l_1 = \frac{wL}{(4\pi R^2)Y}, \Delta l_2 = \frac{wL}{\pi R^2 Y}$$

$$\therefore U = \frac{1}{2}K_1(\Delta l_1)^2 + \frac{1}{2}K_2(\Delta l_2)^2$$

$$= \frac{1}{2} \times \frac{Y\left(4\pi R^2\right)}{L} \times \left[\frac{wL}{4\pi R^2 Y}\right]^2 + \frac{1}{2} \times \frac{Y\left(\pi R^2\right)}{L}$$

$$\times \left[\frac{wL}{\pi R^2 Y}\right]^2 \left(\because K = \frac{YA}{L}\right)$$

$$= \frac{5w^2 L}{8\pi R^2 Y}$$

Q5. Solution

Correct Answer: (C)

Given, area of plates, $A=\pi R^2=3.14\times(0.1)^2\ \mathrm{m}^2$ and $\frac{dE}{dt}=6\times10^{13}\frac{\mathrm{V}}{\mathrm{m}\times\mathrm{s}}$ Displacement current between the plates, $\Rightarrow$

$$\begin{aligned} I_d &= \varepsilon_0\frac{d\phi}{dt} \\ \Rightarrow\quad I_d &= \varepsilon_0 A\left(\frac{dE}{dt}\right) \quad \left[\because \frac{d\phi}{dt}=A\frac{dE}{dt}\right] \\ &= 8.85\times10^{-12}\times3.14\times(0.1)^2\times6\times10^{13} \\ \Rightarrow\quad I_d &= 16.67\ \mathrm{A} \end{aligned}$$

Q6. Solution

Correct Answer: (A)

The changing magnetic field induce an electric field which will accelerate the electron. The acceleration of electron will be $a=\frac{eE}{m}\quad[\because F=qE]$ To find electric field E, we will use Faraday's electromagnetic induction. emf, $\varepsilon=\frac{-d\phi}{dt}(B\cdot A)$ where, B is the magnetic field and A is the area of cross-section.
$\varepsilon=-A\cdot\frac{dB}{dt}=-\pi r^2\frac{dB}{dt}$ Now, electric field, $E=-\frac{\text{emf}}{\text{distance}}=-\frac{\dot{\varepsilon}}{d}=\frac{\pi r^2}{d}\frac{dB}{dt}$ Now, acceleration,

$$\begin{aligned} a=\frac{e}{m}\frac{\pi r^2}{d}\frac{dB}{dt}=\frac{1.6\times10^{-19}}{9.1\times10^{-31}}\times\frac{22}{7}\times\frac{(0.5)^2}{(0.3)}\times50\times10^{-6} \\ =23\times10^6\ \mathrm{ms}^{-2} \end{aligned}$$

Q7. Solution

Correct Answer: (A)

Let the transition of electron in hydrogen atom $(Z=1)$ gives wavelength λ_1 for transition from n_2 to n_1, then $\frac{1}{\lambda}=RZ^2\left(\frac{1}{m^2}-\frac{1}{n^2}\right)$ Here, $\lambda=\lambda_1, Z=1, m=n_1, n=n_2$ $\frac{1}{\lambda_1}=R(1)^2\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)=R\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)$
Now, if electron makes transition in He^+ion from n_4 to n_3 and gives wavelength λ_2, then

$$\begin{aligned} \frac{1}{\lambda_2} &= R(2)^2\left(\frac{1}{n_3^2}-\frac{1}{n_4^4}\right) \\ &= R\left[\frac{1}{\left(\frac{n_3}{2}\right)^2}-\frac{1}{\left(\frac{n_4}{2}\right)^2}\right] \end{aligned}$$

Given that, $\lambda_1=\lambda_2\Rightarrow R\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)=R\left[\frac{1}{\left(\frac{n_3}{2}\right)^2}-\frac{1}{\left(\frac{n_4}{2}\right)^2}\right]$ On comparing beth sides, we get $n_1=\frac{n_3}{2}$ and $n_2=\frac{n_4}{2}$ The value of n_1, n_2, n_3 and n_4 must be an integer, as they denote principal quantum number. If $n_3=2$ and $n_4=4$, then the given condition is satisfied.

Q8. Solution

Correct Answer: (D)

Power of equiconvex lens $=P$ i.e $P=\frac{1}{f}$, where f is focal length of equiconvex lens. When an equiconvex lens is cut into two symmetrical halves by a plane containing the principal axis, then its focal length remains same.

Hence, power of lens (each halves)

$$\begin{aligned} P' &= \frac{1}{f}\Rightarrow P'=P \\ \Rightarrow\quad \frac{P'}{P} &= 1 \\ \Rightarrow\quad P:P &= 1:1 \end{aligned}$$

Q9. Solution

Correct Answer: (B)

Given, path difference, $\Delta x = \frac{\lambda}{4}$ Phase difference, $\phi = \left(\frac{2\pi}{\lambda}\right) \times$ (Path difference) $= \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2} \ldots (i)$ It is given that, maximum intensity is I_0. So, intensity at a point on the screen, where phase difference ϕ is given by

$$I = I_0 \cos^2\left(\frac{\phi}{2}\right)$$

$$= I_0 \times \cos^2\left(\frac{\pi/2}{2}\right) \quad [\because \text{ using Eq.(i) }]$$

$$= I_0 \times \left(\frac{1}{\sqrt{2}}\right)^2$$

$$= \frac{I_0}{2}$$

Q10. Solution

Correct Answer: (C)

Magnifying power of telescope,m = 9 $\Rightarrow \frac{f_0}{f_e} = 9$ $\Rightarrow f_0 = 9f_e \ldots (i)$ Distance between objective and eyepiece is given as $f_0 + f_e = 20 \ldots (ii)$

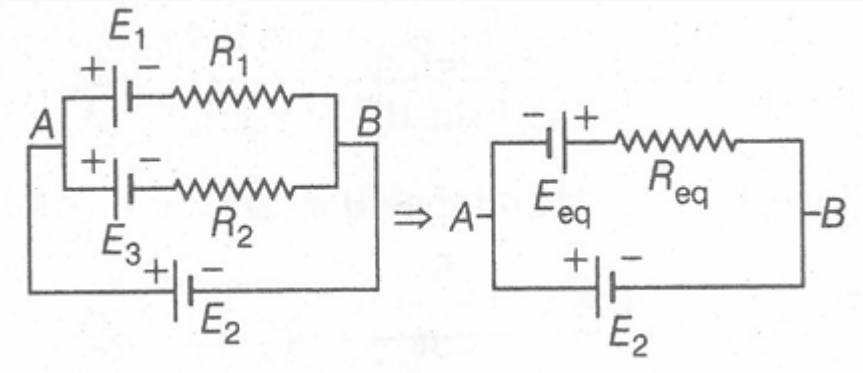
$\Rightarrow 9f_e + f_e = 20$ [from Eq. (i)] $\Rightarrow 10f_e = 20$ $\Rightarrow f_e = 2$ cm $\therefore$ From Eq. (i), $f_0 = 9 \times 2 = 18$ cm

$\therefore \frac{f_0}{f_e} = \frac{18}{2} = 9$,

Q11. Solution

Correct Answer: (B)

The given circuit can be redrawn as,



where, equivalent emf of the combination, $E_{\text{eq}} = \frac{E_1R_1 + E_2R_2}{R_1 + R_2} = \frac{2 \times 4 + 2 \times 4}{4 + 4} = 2$ V and

$R_{eq} = \frac{R_1R_2}{R_1+R_2} = \frac{4\times4}{4+4} = \frac{16}{8} = 2\Omega$ Net emf of the circuit, $E' = E_2 + E_{\text{eq}} = 2 + 2 = 4$ V $\therefore$ Current flowing through branch AB, $I = \frac{E'}{R_{eq}} = \frac{4}{2} = 2$ A Clearly from figure, it is clear that direction of current is from point A to B clockwise is loop. ~

Q12. Solution

Correct Answer: (B)

In the given cycle, heat is supplied in process D to A at constant volume and in process A to B at constant pressure. $\therefore$ Heat supplied, $Q = nC_V(\Delta T)_{DA} + nC_p(\Delta T)_{AB}$ For an ideal monoatomic gas,

$$C_V = \frac{3}{2}R \text{ and } C_p = \frac{5}{2}R$$

$$\Rightarrow \quad Q = \frac{3}{2}nR\Delta T + \frac{5}{2}nR\Delta T \quad [\because nR\Delta T = p\Delta V]$$

$$= \frac{3}{2}(p_0V_0) + 5\,(p_0V_0)$$

$$= \frac{13}{2}p_0V_0$$

Q13. Solution

Correct Answer: (C)

$E = 3.65 \times 10^4 \text{ N/C}, r = 1\mu \text{ m} = 10^{-6} \text{ m}$ $\rho_{\text{oil}} = 1.26 \text{ g/cm}^3 = 1.26 \times 10^3 \text{ kg/m}^3$ Since, droplet is stationary hence weight of droplet $=$ force due to electric field $\Rightarrow$ $\frac{4}{3}\pi r^3 \cdot \rho_{\text{oil}} \cdot g = qE \ldots (i)$ If n be the number of excess electrons in the oil drop, then $q = ne$ Hence, from Eq. (i), $\frac{4}{3}\pi r^3 \rho_{\text{oil}} \cdot g = neE$

$$\Rightarrow n = \frac{4\pi r^3 \rho_{\text{oill}}\, g}{3eE}$$

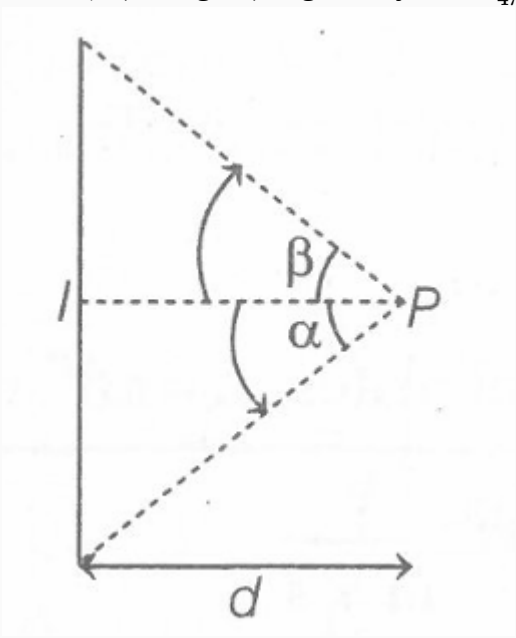
$$= \frac{4 \times 3.14 \times \left(10^{-6}\right)^3 \times 1.26 \times 10^3 \times 10}{3 \times 1.6 \times 10^{-19} \times 3.65 \times 10^4},$$

$$= 9.03 \simeq 9$$

Q14. Solution

Correct Answer: (A)

The magnetic field at a point P due to a current carrying straight wire, such that the ends of the wires, makes angle α and β (see figure) is given by $B = \frac{\mu_0 I}{4\pi d}(\sin\alpha - \sin\beta)$ where, d is the distance between the point P and wire.



In the given problem, $\alpha = \theta_1$ and $\beta = \theta_2 \therefore \quad B = \frac{\mu_0 I}{4\pi d}(\sin\theta_1 - \sin\theta_2)$

Q15. Solution

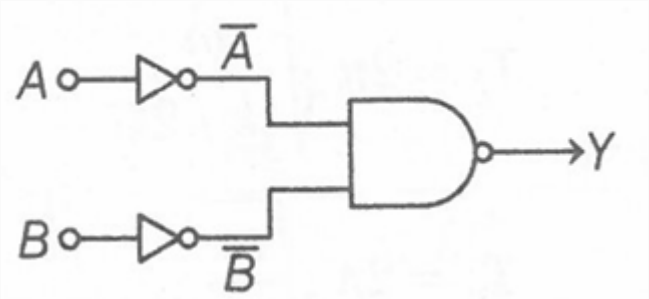
Correct Answer: (B)

Position of first bright fringe, $y_1 = \frac{\lambda D}{d}$ Position of second bright fringe, $$\begin{aligned} y_2 &= \frac{2\lambda D}{d} \\ y_2 - y_1 &= \frac{2\lambda D}{d} - \frac{\lambda D}{d} \\ y_2 - y_1 &= \frac{\lambda D}{d} \\ or \lambda &= \frac{(y_2 - y_1)d}{D} \end{aligned}$$ Substituting given values, we have $$\begin{aligned} \lambda &= \frac{0.553 \times 10^{-3} \times 2.08 \times 10^{-3}}{1.8} \\ &= 639 \times 10^{-9} \text{ m} \\ or \lambda &= 639 \text{ nm} \end{aligned}$$.

Q16. Solution

Correct Answer: (D)

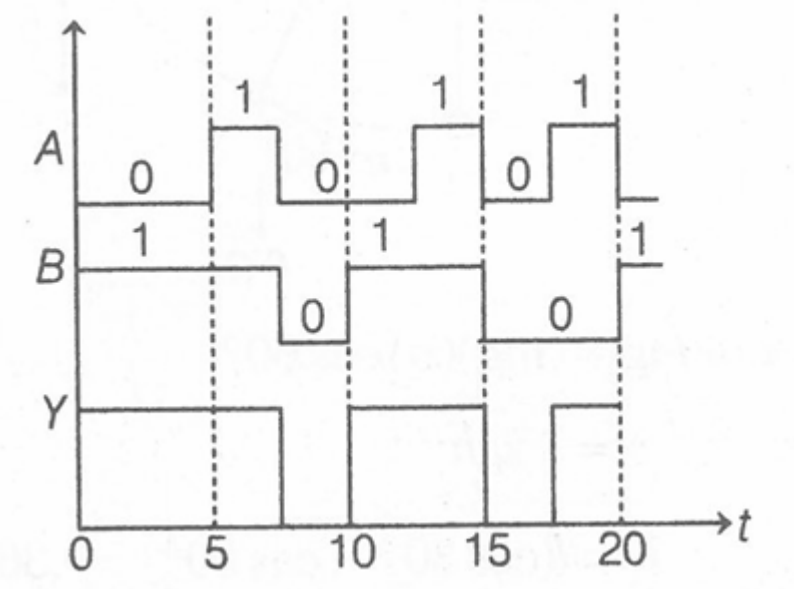
Given circuit can be shown as below



A	B	Y=A+B
0	0	0
0	1	1
1	0	1
1	1	1

Output, $Y = \overline{(\bar{A})(\bar{B})} = \bar{A} + \bar{B} = A + B$ The truth table will be

The corresponding input and output waveforms are shown below.



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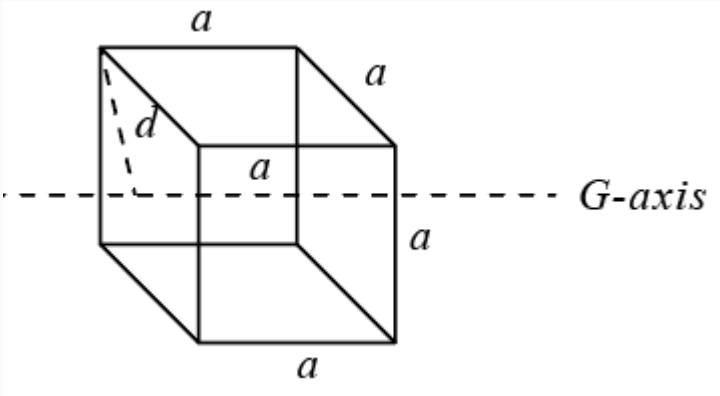
Q17. Solution

Correct Answer: (A)



From theorem of perpendicular axes, we have

$$I = I_C + m\left(\frac{a}{\sqrt{2}}\right)^2$$

$$= \left[\frac{ma^2}{12} + \frac{ma^2}{12}\right] + \frac{ma^2}{2} \wedge$$

$$= \frac{2}{3}ma^2$$

Q18. Solution

Correct Answer: (C)

Volume of eight drops = Volume of a big drop $\therefore \quad \left(\frac{4}{3}\pi r^3\right) \times 8 = \frac{4}{3}\pi R^3 \Rightarrow \quad 2r = R$...(i) According to charge conservation, $8q = Q$...(ii) Potential of one small drop, $V' = \frac{q}{4\pi\varepsilon_0 r}$ Similarly, potential of big drop, $V = \frac{Q}{4\pi\varepsilon_0 R}$ Now, $\frac{V'}{V} = \frac{q}{Q} \times \frac{R}{r} \Rightarrow \quad \frac{V'}{20} = \frac{q}{8q} \times \frac{2r}{r}$ [from Eqs. (i) and (ii)] $\therefore \quad V' = 5$ V ^

Q19. Solution

Correct Answer: (A)

$$T_a = 2\pi\sqrt{\frac{m}{k}}$$

$$T_b = 2\pi\sqrt{\frac{m}{(k/2)}}$$

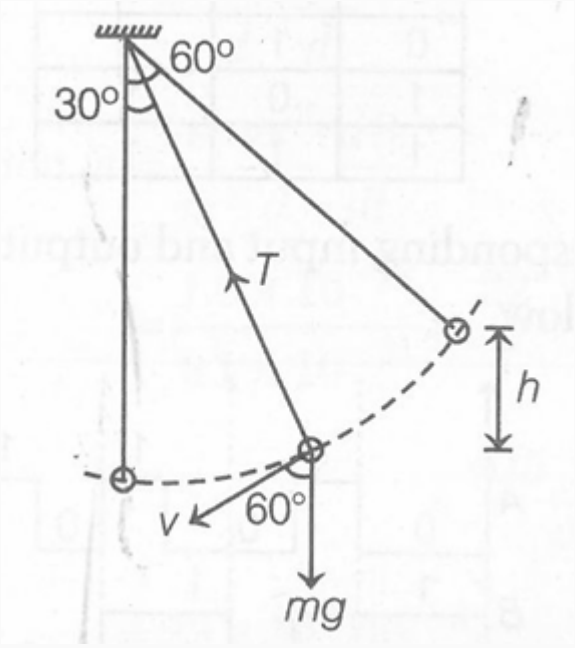
!

$$T_c = 2\pi\sqrt{\frac{m}{2k}}$$

$$\therefore \quad T_a : T_b : T_c = 1 : \sqrt{2} : \frac{1}{\sqrt{2}}$$

Q20. Solution

Correct Answer: (A)



Power of tension $= 0$

Power of $mg = (mg)(v)\cos 60^\circ$

Here, $\quad v = \sqrt{2gh}$

and $\quad h = l\left(\cos 30^\circ - \cos 60^\circ\right) = 0.36 \text{ m}$

$v = \sqrt{2 \times 10 \times 0.36}$

$= 2.68 \text{ m/s}$,

Power $= (mg)(v)\cos 60^\circ$

$= (1 \times 10)(2.68) \times \dfrac{1}{2}$

$= 1.34 \times 10 = 13.4 \text{ W}$

Q21. Solution

Correct Answer: (A)

At maximum height, inclination with horizontal becomes zero. Time (taken by) projectile to reach maximum height,

$$t = \frac{u \sin\theta}{g}$$

$$\Rightarrow \quad u = \frac{gt}{\sin\theta} \ldots (i)$$

Since, $R = u\cos\theta \times (2t)$

$$\Rightarrow \quad \cos\theta = \frac{R}{2ut}$$

$$\Rightarrow \quad \cos\theta = \frac{R}{2 \cdot \frac{gt}{\sin\theta} \cdot t}$$

$$= \frac{R\sin\theta}{2gt^2} (Using(i))$$

$$\Rightarrow \quad \cot\theta = \frac{R}{2gt^2} = \frac{R}{2 \times 10t^2} = \frac{R}{20t^2}$$

Q22. Solution

Correct Answer: (C)

Mass of dust particle, $m = 4 \times 10^{-12}$mg $= 4 \times 10^{-12} \times 10^{-3}$ g $= 4 \times 10^{-12} \times 10^{-3} \times 10^{-3}$ kg Electric field, $E = 50$ N/C Weight of dust particle, $W = mg = 4 \times 10^{-18} \times 10 = 4 \times 10^{-17}$ N Electric force experienced by dust particle, $F_e = qE$ $F_e = ne \cdot E = n \times 1.6 \times 10^{-19} \times 50$ where, n is the number of electrons removed from neutral dust particle.

$$\text{Electric force} = \text{Weight of dust particle}$$

$$n \times 1.6 \times 10^{-19} \times 50 = 4 \times 10^{-17}$$

$\therefore$ At balance condition, $n = \dfrac{4 \times 10^{-17}}{1.6 \times 10^{-19} \times 50}$

$$= \frac{400}{80} = 5$$

Q23. Solution

Correct Answer: (A)

$F = kx \therefore \quad k = \frac{F}{x} = \frac{100}{1}$ N/m $= 100$ N/m Now, from energy conservation, between natural length of spring and its maximum compression state. $\frac{1}{2}mv^2 + mgh = \frac{1}{2}kx_{\max}^2$

$$\therefore \quad v = \sqrt{\frac{kx_{\max}^2}{m} - 2gh}$$

$$= \sqrt{\frac{(100)(2)^2}{10} - (2)(10)\left(\frac{2}{2}\right)} = \sqrt{20} \text{ m/s}$$

Q24. Solution

Correct Answer: (D)

Given, $A = 1.0$ m ± 0.2 m, $B = 2.0$ m ± 0.2 m Let $Y = \sqrt{AB} = \sqrt{(1.0)(20)} = 1.414$ m Rounding off to two significant digits $Y = 1.4$ m

$$\frac{\Delta Y}{Y} = \frac{1}{2}\left[\frac{\Delta A}{A} + \frac{\Delta B}{B}\right]$$

$$= \frac{1}{2}\left[\frac{0.2}{1.0} + \frac{0.2}{2.0}\right]$$

$$= \frac{0.6}{2 \times 20}$$

Rounding off to one significant digit,

$$\Rightarrow \quad \Delta Y = \frac{0.6Y}{2 \times 20} = \frac{0.6 \times 1.4}{2 \times 20} = 0.212$$

$\Delta Y = 0.2$ m

Q25. Solution

Correct Answer: (A)

$\omega = \alpha - \beta t$. Comparing with $\omega = \omega_0 - \alpha t$ Initial angular velocity $= \alpha$ Angular'retardation $= \beta \therefore$ Angle rotated before it stops is $\frac{\alpha^2}{2\beta}$. $\left[\left[\text{using } \omega^2 = \omega_0^2 + 2\beta\theta\right]\right.$

Q26. Solution

Correct Answer: (C)

$E = \frac{Kq}{r^2} \cdot \hat{r} = \frac{Kq}{r^3} \cdot r$ Here,

$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$
$$= \sqrt{(30)^2 + (30)^2 + 0^2}$$
$$= 30\sqrt{2} \text{ cm}$$
$$= 30\sqrt{2} \times 10^{-2} \text{ m}$$

and $q = 8 \times 10^{-3} \times 10^{-6}$C Also,

$$\text{So, E} = \frac{9 \times 10^9 \times 8 \times 10^{-3} \times 10^{-6}}{\left(30\sqrt{2} \times 10^{-2}\right)^3} \times (30\hat{\mathbf{i}} + 30\hat{\mathbf{j}}) \times 10^{-2}$$

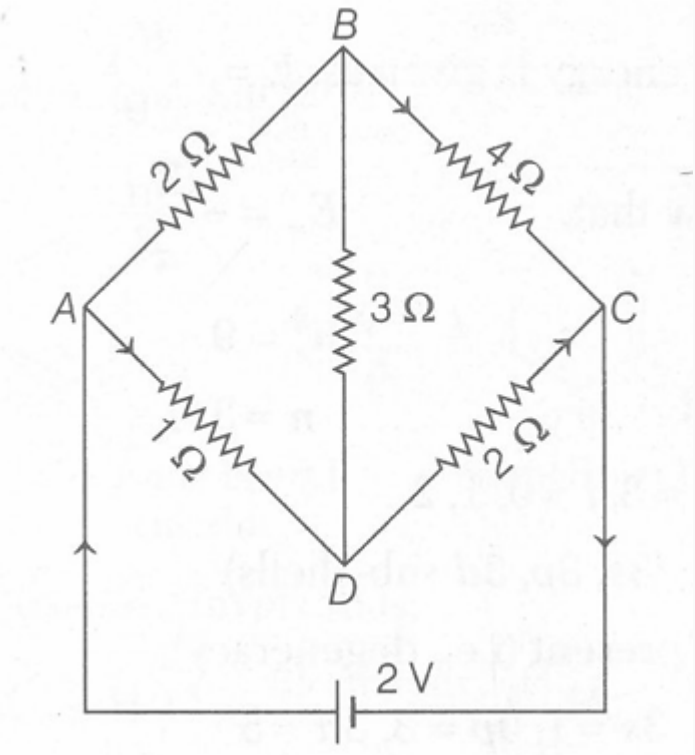
$\mathbf{r} = (30\hat{\mathbf{i}} + 30\hat{\mathbf{j}}) \times 10^{-2}$ m

$$= \frac{9 \times 8 \times 10^9 \times 10^{-11}}{27 \times 2\sqrt{2} \times 10^{-6} \times 10^3} \times 30(\hat{\mathbf{i}} + \hat{\mathbf{j}})$$
$$= \frac{9 \times 8 \times 10^2 \times 3}{27 \times 2\sqrt{2}}(\hat{\mathbf{i}} + \hat{\mathbf{j}})$$
$$= 200\sqrt{2}(\hat{\mathbf{i}} + \hat{\mathbf{j}})\text{NC}^{-1}$$

Q27. Solution

Correct Answer: (C)

The equivalent Wheatstone's bridge network of the given circuit is shown in figure. Here,



The points B and D are at the same potential as the bridge is balanced. So, the 3Ω resistance in BD arm is ineffective and can be omitted from the circuit. Hence, equivalent resistance remains same when 3Ω resistor is removed.

Q28. Solution

Correct Answer: (C)

On stretching, volume of wire remains constant. So, $V = Al$ or $l = \frac{V}{A}$

$\therefore \quad R = \rho \cdot \frac{l}{A} = \frac{\rho V}{A^2} = \frac{\rho V}{\frac{\pi^2 D^4}{16}} = \frac{16 \rho V}{\pi^2 D^4}$

$\therefore \quad \frac{\Delta R}{R} = -4 \frac{\Delta D}{D} = -4(-0.4) = 1.6\%$

Q29. Solution

Correct Answer: (B)

Velocity on hitting the surface $= \sqrt{2 \times 9.8 \times 4.9}$ $= 9.8 \text{ m/s}$ Velocity after first bounce, $v = \frac{3}{4} \times 9.8$ Time taken from first bounce to the second bounce $= \frac{2v}{g} = 2 \times \frac{3}{4} \times 9.8 \times \frac{1}{9.8} = 1.5 \text{ s}$

Q30. Solution

Correct Answer: (C)

$(v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$ and $v_{\text{sound}} = \sqrt{\frac{\gamma RT}{M}}$ $\frac{v_{\text{sound}}}{v_{\text{rms}}} = \sqrt{\frac{\gamma}{3}}$ Degree of freedom is 6 .

$\therefore \quad \gamma = 1 + \frac{2}{f} = 1 + \frac{2}{6} = \frac{4}{3}$

$\therefore \quad v_{\text{sound}} = \sqrt{\frac{4/3}{3}} v_{\text{rms}} = \frac{2}{3} v_{\text{rms}} = \frac{2c}{3}$

Q31. Solution

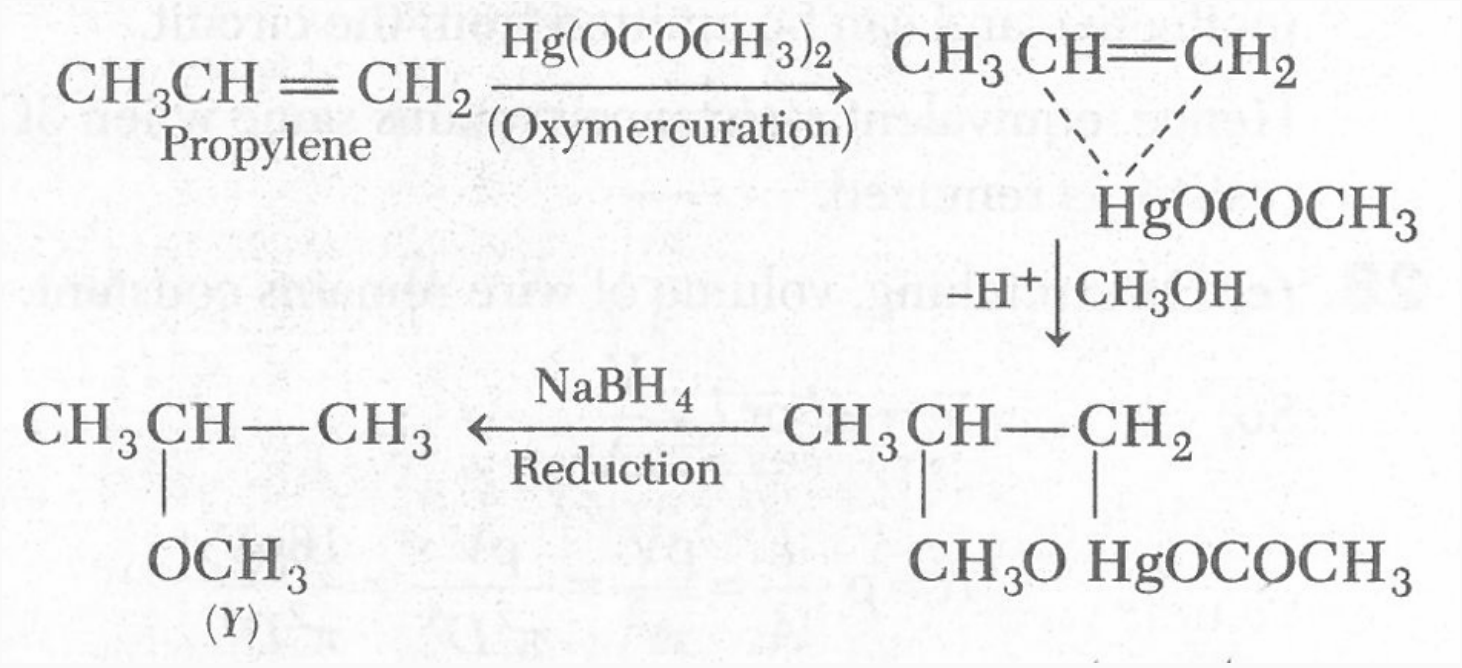
Correct Answer: (D)

The energy is given as, $E = -\frac{R_{\text{H}}}{9} \ldots (i)$ We know that, $E_n = -\frac{R_{\text{H}}}{n^2} \ldots (ii)$ $n^2 = 9$ $n = 3$ When $n = 3, l = 0, 1, 2$ $(3s, 3p, 3d$ sub-shells$)$ Orbitals present (i.e., degeneracy), $3s = 1, 3p = 3, 3d = 5$ Total number of orbital present $= 3 + 1 + 5 = 9$

Q32. Solution

Correct Answer: (B)

The complete reaction is as follows,



Q33. Solution

Correct Answer: (A)

$\alpha = 0.5$ (50% dissociated) The reaction involves is

	$K_4[Fe(CN)_6]$	$\rightleftharpoons$	$4K^+ + [Fe(CN)_6]^4$	
Initial moles	1	0	0	Total moles at equilibrium
Moles after dissociation	$1-\alpha$	4α	α	

$= 1 - \alpha + 4\alpha + \alpha$ Total $= 1 + 4\alpha \therefore$ van't Hoff factor $(i) = \frac{1+4\alpha}{1} = 1 + 4 \times 0.5 = 3$ Osmotic pressure,

$C = 0.1\text{M} = 10^2 \text{ mol m}^{-3}$

$\pi = iCRT$ $\pi = 3 \times 10^2 \times 8.314 \times 300$ or $\pi = 7.48$ atm

$= 7.483 \times 10^5 \text{Nm}^{-2}$

Q34. Solution

Correct Answer: (D)

In a cyclic process, internal energy change $= 0$ According to 1st law of thermodynamics $\Delta U = \Delta q + \Delta W$ As, $\Delta U = 0, \quad \Delta q = -\Delta W$ From the graph, work done $= \Delta W =$ Area under curve Here, area of circle $= \pi r^2$

$$= \pi\left(\frac{25-5}{2}\right)^2 = \pi \times 10^2$$

$\therefore \quad \Delta q = 100\pi$ J

Q35. Solution

Correct Answer: (A)

Oxidation half reaction: $SO_2(\text{ g}) + 2H_2O(l) \longrightarrow HSO_4^-(aq) + 3H^+(aq) + 2e^- \ldots (i)$ Reduction half reaction: $MnO_4^-(aq) + 8H^+ + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l) \ldots (ii)$ Multiply eq. (i) by 5 and (ii) by 2 and then add $\begin{aligned} &2MnO_4^-(aq) + 5SO_2(g) + 2H_2O(l) + H^+(aq) \\ &\longrightarrow 2Mn^+(aq) + 5HSO_4^-(aq) \end{aligned}$ Thus, the stoichiometric coefficient of SO_2 in the above balance equation is 5 .

Q36. Solution

Correct Answer: (B)

Let the wavelength of given particle $= x$ Velocity $(v) = \frac{x}{100}$ Using wavelength formula

$$\lambda = \frac{h}{mv}$$

$$\therefore x = \frac{h}{m\left(\frac{x}{100}\right)}$$

$$x = \frac{100h}{mx}$$

$$x^2 = 100\frac{h}{m}$$

$$\Rightarrow \quad x = 10\sqrt{\frac{h}{m}}$$

Q37. Solution

Correct Answer: (B)

When O^-changes into O^{2-}, energy is absorbed i.e the change is endothermic because O^-repels the incoming electron due to similar charge. H ice, it needs energy to add (accept) the electron. So, electron affinity (EA_2) is positive.

Q38. Solution

Correct Answer: (D)

Let the bond dissociation energy of $X_2 = a$ kJ/mol i.e. BE $(X_2) = a$ kJ/mol Then, BE $(Y_2) = 0.5a$ and BE$(XY) = a$ kJ/mol Given, $\frac{1}{2}X_2 + \frac{1}{2}Y_2 \longrightarrow XY, \Delta H = -200$ kJ/mol

$$\Delta_r H = \text{BE(Rreactants)} - \text{BE(Product)}$$

$$\Delta_r H = \left[\frac{1}{2}\text{BE}\left(X_2\right) + \frac{1}{2}\text{BE}\left(Y_2\right)\right] - \text{BE}(XY)$$

$$\therefore -200 = \frac{a}{2} + \frac{0.5a}{2} - a$$

$$a = \frac{200}{0.25} = 800 \text{ kJ/mol}$$

Q39. Solution

Correct Answer: (C)

Stronger the acid, lower is the pH or weaker the acid, higher the pH . The strength of the given acid is in the order $\mathrm{HCl} > \mathrm{HCOOH} > \mathrm{CH_3COOH}$ Hence, their pH is in the order $\mathrm{CH_3COOH} > \mathrm{HCOOH} > \mathrm{HCl}$

Q40. Solution

Correct Answer: (A)

$\mathrm{N}^{3-}, \mathrm{O}^{2-}$ and $\mathrm{F}^{-}$are isoelectronics (10 electrons) species. Their ionic radii will decrease with increase in magnitude of the nuclear charge from $\mathrm{N}(\mathrm{Z}=7)$ to O $(Z=8)$ and $F(Z=9)$. i.e.

$$\begin{array}{ccc} \mathrm{N}^{3-} > & \mathrm{O}^{2-} > & \mathrm{F}^{-} \\ 1.71\tilde{A}\ldots & 1.40\tilde{A}\ldots & 1.36\tilde{A}\ldots \end{array}$$

Hence, option (1) is correct.

Q41. Solution

Correct Answer: (B)

For an acidic buffer, $\mathrm{pH}=\mathrm{p}K_{\alpha}+\log \frac{[\text{ Salt }]}{[\text{ Acid }]} \ldots(i)$ Let the concentration of potassium acetate solution $=x\mathrm{M}$

20 mL of 0.1 M acetic acid $=20 \times 0.1$millimol $=2$ millimol 50 mL of $x\mathrm{M}$ potassium acetate

$$\begin{aligned} &= x \times 50\text{millimol} \\ &= 50x \text{ milimol} \\ &\mathrm{pH}=4.8(Given) \end{aligned}$$

Substitute all value in eq. (i),

$$4.8=-\left(\log 1.8+10^{-5}\right)+\log \frac{50x}{2}$$

$$\log 25x=0.06$$

$$x=0.04\mathrm{M}$$

Q42. Solution

Correct Answer: (A)

$$\begin{aligned} 58.5 \text{ gNaCl} &= \frac{58.5}{58.5} \text{ mol} \\ &= 1 \text{ mol} \\ 180 \text{ g of glucose } &= \frac{180}{180}=1 \text{ mol} \end{aligned}$$

Thus, we have 1 M NaCl and 1 M glucose solution. But NaCl dissociates in the solution and glucose does not dissociates. Thus particle concentration is more in NaCl . Hence, NaCl solution will have higher elevation in boiling point.

Q43. Solution

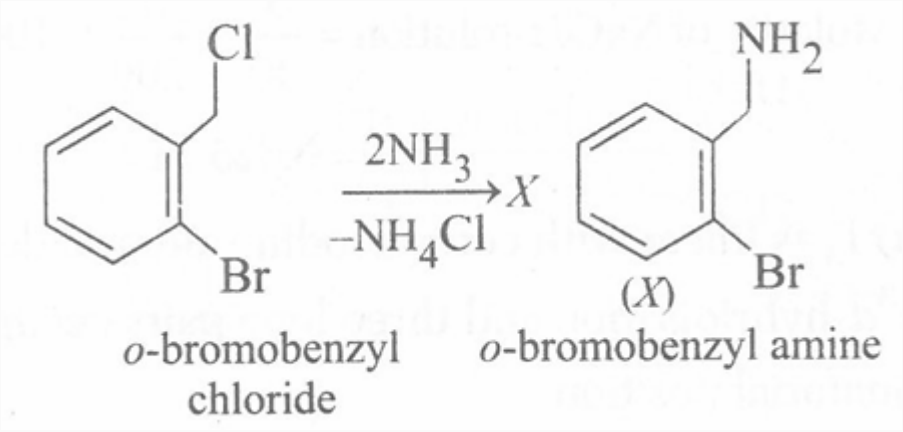
Correct Answer: (A)

Although $\mathrm{CH_3C}=\mathrm{CH}-\mathrm{CH}=\mathrm{CH_2}$ has two double bond but one of them has two similar atoms (i.e. H - atom) therefore no geometrical isomerism is possible around the double bond. However each carbon atom of other double bond has two different atoms/groups and hence geometrical isomerism is possible around this double bond. Hence, only two geometrical isomers are possible.

Q44. Solution

Correct Answer: (C)

Since, benzyl halides are more reactive than aryl halides, therefore reaction occur at the benzyl chloride part. The reaction is as follow,



Q45. Solution

Correct Answer: (D)

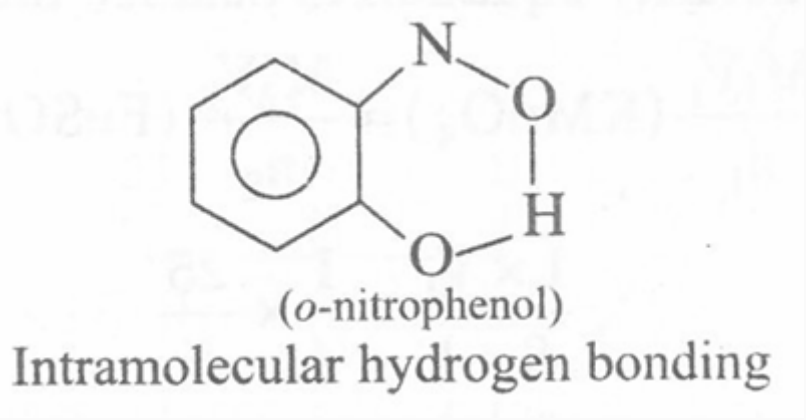
The reaction involved is, $\mathrm{NaCl} + \mathrm{H_2O} \xrightarrow{\text{Electrolysis}} \frac{1}{2}\mathrm{H_2} + \mathrm{Cl_2} + \mathrm{NaOH}$ Amount of NaCl present in 0.5 L of $1.0\mathrm{M}, \mathrm{NaCl} = 0.5$ mol Quantity of electricity passed $= 965 \times 5$ coulombs. $\therefore$ 4825 coulombs will decompose NaCl . $= \frac{4825}{96500}$ mol $= 0.05$ mol NaOH formed in the solution will also be 0.05 mol . Volume of solution $= 0.5\ \mathrm{L} \therefore$

Molarity of NaOH in the solution $= \frac{0.05\text{moles}}{0.5\ \mathrm{L}} = 0.1\mathrm{M} = 10^{-1}\mathrm{M}$ $\therefore$ $\mathrm{pOH} = 1$ or $\mathrm{pH} = 14 - 1 = 13$

Q46. Solution

Correct Answer: (A)

In o-nitrophenol, hydrogen of $\mathrm{O} - \mathrm{H}$ group and oxygen of nitro group form intramolecular hydrogen bonding producing a six membered ring structure while in *m*-nitrophenol, *p*-nitrophenol and phenol, intermolecular hydrogen bonding are present. The structure is,



Q47. Solution

Correct Answer: (A)

The overall reaction of $H_2 - O_2$ fuel is, $2H_2 + O_2 \longrightarrow 2H_2O$ Thus $4\,F = 4 \times 96500C$ produce water $= 2$ mol $= 36$ g

$$\text{Charge actually passed } = i \times t$$
$$= 1 \times 595.1 \times 60 \times 60$$
$$= 2142360C$$

$$\therefore \text{ Water produced } = \frac{36}{4 \times 96500} \times 2142360$$

$\cong 200$ g or 200 mL

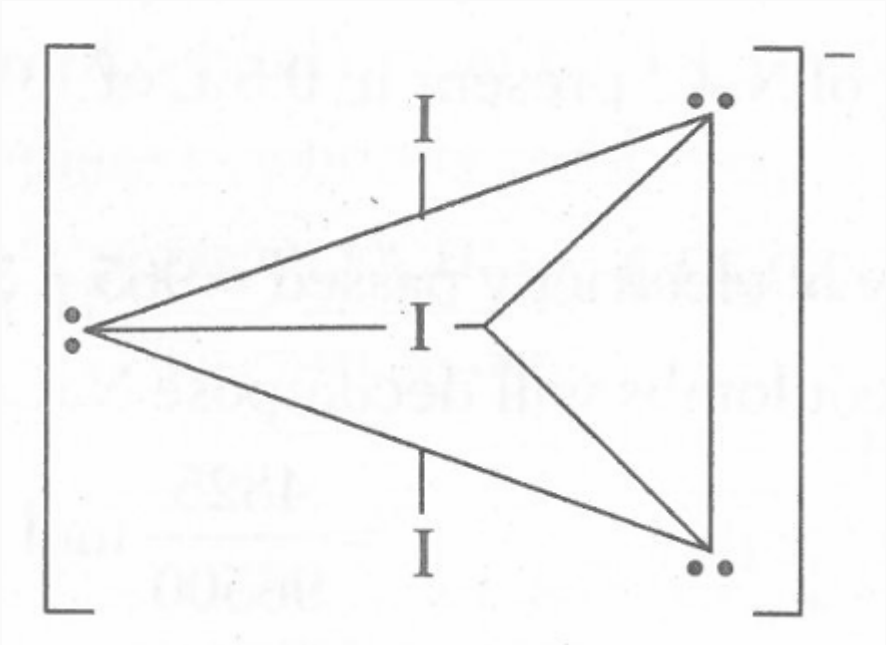
$$\therefore \text{ Molarity of NaOH solution } = \frac{5}{40} \times \frac{1}{200} \times 1000$$

$= 0.625M$

Q48. Solution

Correct Answer: (A)

I_3^- is linear with central iodine atom undergoing sp^3d-hybridisation and three lone pairs occupying equatorial position. Thus, this structure include d-orbital contribution to the hybridisation scheme. The structure is,



PCl_3 structure is sp^3-hybridised with one lone pair. NO_3^- is sp^2-hybridised and H_2Se is sp^3 - hybridised with two lone pairs of Se atom.

Q49. Solution

Correct Answer: (D)

Aldehydes which contain α-hydrogen on a saturated carbon, i.e. CH_3CH_2CHO will undergo aldol condensation. Although $CH_2 = \overset{\alpha}{C}\,H - CHO$ also contain a α-hydrogen atom but cannot be easily abstracted by a base to form a carbanion and hence does not undergo aldol condensation.

Q50. Solution

Correct Answer: (D)

The balance ionic equation for the reaction is $MnO_4^- + 5Fe^{2+} + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$ From the above equation it is clear that 1 mole of $KMnO_4 = 5$ moles of $FeSO_4$. Apply molarity equation to balance the redox reaction

$$\frac{M_1V_1}{n_1}(KMnO_4) = \frac{M_2V_2}{n_2}(FeSO_4)$$

$$\frac{1 \times V_1}{8 \times 1} = \frac{1}{4} \times \frac{25}{5}$$

$$V_1 = \frac{1 \times 25 \times 8}{4 \times 5}$$

$$V_1 = 10 \text{ cm}^3 \text{ or } V_1 = 10.0 \text{ mL}$$

Q51. Solution

Correct Answer: (C)

Deficiency of riboflavin (vitamin B_2) can cause cheilosis and digestive disorder.

Q52. Solution

Correct Answer: (C)

The rate of S_n2 reaction is the maximum with 1° allyl halide having least steric hindrance. Out of CH_3CH_2Br and $CH_3CH_2CH_2Br$, CH_3CH_2Br has the least steric hindrance and hence has the highest relative rate.

Q53. Solution

Correct Answer: (B)

$2SO_2 + O_2 \rightleftharpoons 2SO_3$ Rate of reaction $= -\frac{1}{2}\frac{d[SO_2]}{dt} = -\frac{d[O_2]}{\text{dt}} = \frac{1}{2}\frac{d[SO_3]}{dt}$ Rate of appearance of SO_3,

$$\frac{d\left[SO_3\right]}{dt} = 2 \times \left[\frac{-d\left[O_2\right]}{dt}\right] = 2 \times \left(2 \times 10^{-4}\right)$$

$$= 4 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

Q54. Solution

Correct Answer: (D)

According to Arrhenius equation, $\log k = \log A - \frac{E_a}{2.303RT} \ldots (i)$ For 1st order reaction, $k = \frac{0.693}{t_{1/2}} = \frac{0.693}{600s} = 1.1 \times 10^{-3} \text{ s}^{-1}$ Substitute the value of k in Eq. (i) $log\left(1.1 \times 10^{-3}\right) = \log\left(4 \times 10^{13}\right) - \frac{98.6 \times 10^3}{2.303 \times 8.314 \times T}$ Solving for T, $T = 311.15$ K

Q55. Solution

Correct Answer: (A)

Since glycine (gly) is achiral, alanine (Ala) is the C-terminal amino acid, therefore N -terminal amino acid can be valine (val) or serine (ser). In the view of these condition, the following four sequences for the tetrapeptide is possible. (i) Val-gly-Ser-Ala (ii) Val-Ser-gly-Ala (iii) Ser-gly-Val-Ala (iv) Ser-Val-gly-Ala

Q56. Solution

Correct Answer: (C)

Among the given species, Cr^{2+} and Mn^{3+} have same number of unpaired electrons.

Unpaired electrons, $n = 4$

Magnetic moment, $\mu = \sqrt{n(n+2)} = \sqrt{4(4+2)}$
$= \sqrt{24}\text{BM}$

Q57. Solution

Correct Answer: (C)

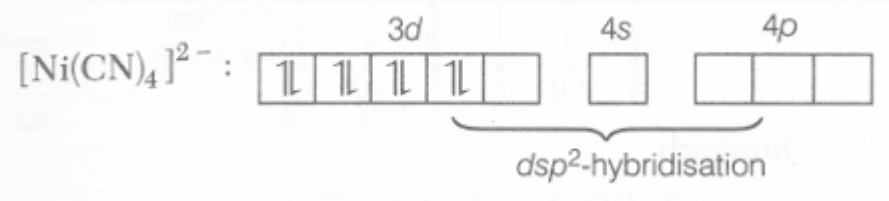
Electronic configuration of Fe . $_{26}\text{Fe} = [\text{Ar}]3d^6 4s^2$
$\text{Fe}^{3+} = [\text{Ar}]3d^5$ Number of unpaired electrons, $n = 5$ Spin only magnetic moment, $\mu_s = \sqrt{n(n+2)}\text{BM}$

$$\mu_s = \sqrt{5(5+2)}$$
$$\mu_s = \sqrt{35}$$
$$\mu_s = 5.85\text{BM or } \mu_s \simeq 6\text{BM}$$

Q58. Solution

Correct Answer: (D)

In $[\text{Ni}(\text{CN})_4]^{2-}$, $\text{Ni}^{2+} = 3d^8 4s^0$ $\text{AsCN}^-$is a strong ligand, pairing of electrons take place, Therefore,



dsp^2 hybridisation gives square planar geometry.

Q59. Solution

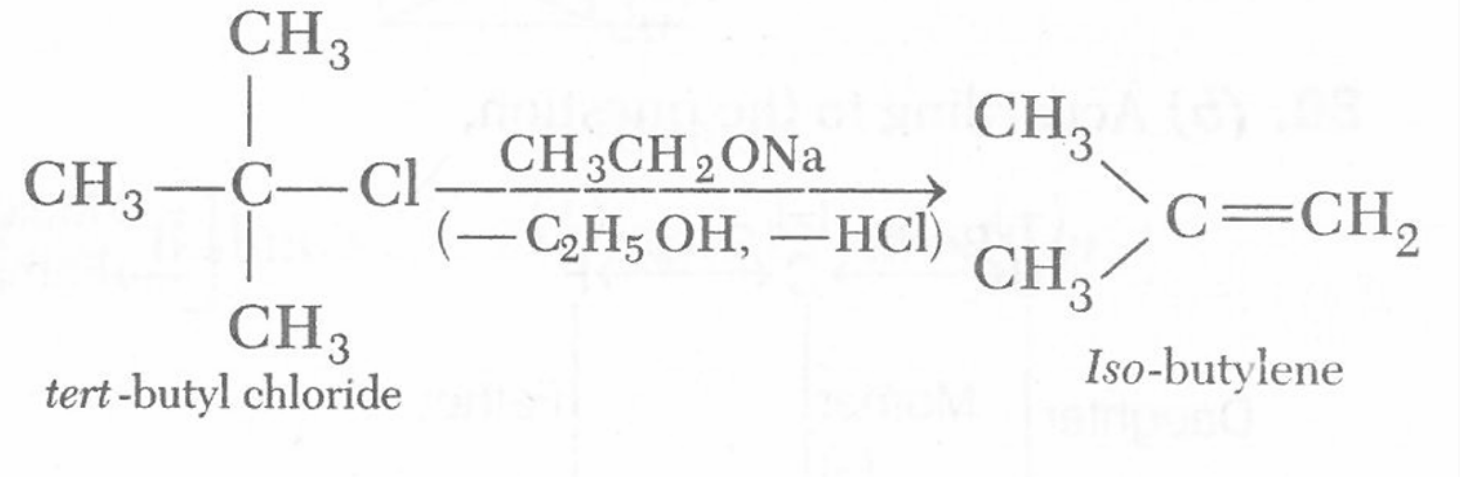
Correct Answer: (C)

(A) $K_3\,[CO\,(C_2O_4)Cl_2]$
$\Rightarrow [CO(C_2O_4)_2Cl_2]^{3-}$
$\Rightarrow m - 2(2) - 2 = -3$
$m = 3$
(B) $[Pt\,(C_2H_4)Cl_3]^{-1}$
$m + 0 - 3 = -1$
$m = +2$
(C) $[Fe(H_2O)_5(NO)]^{2+}$
$m + 1 = 2$
$m = 1$
(D) $[Ni(CO)_4]$
$= m + 0 = 0$
$m = 0$

Q60. Solution

Correct Answer: (B)

Tertiary alkyl halides readily undergo dehydrohalogenation to form alkenes.



Q61. Solution

Correct Answer: (B)

The underlined word 'fortitude' means courage shown when you are in great pain or experiencing a lot of trouble. Hence, 'bravery' from the given options is its synonym.

Q62. Solution

Correct Answer: (D)

The underlined word 'dissipate' means to disappear gradually.

Q63. Solution

Correct Answer: (A)

The passive form of the given sentence is 'The poor people in your parish are to be helped by you'.

Q64. Solution

Correct Answer: (D)

The underlined word 'jeopardy' means danger of loss, harm or failure. Hence, its antonym is 'safety'.

Q65. Solution

Correct Answer: (B)

The underlined word 'cacophony' means a harsh discordant mixture of sounds. Hence, its antonym is 'symphony' which means consonance of sounds.

Q66. Solution
Correct Answer: (A)

A formal statement testifying to someone's character and qualifications is called a 'Testimonial'.

Q67. Solution
Correct Answer: (D)

The emblems or insignia of royalty, especially the crown, sceptre and other ornaments used at a coronation or the distinctive clothing worn as an indication of status is called Regalia.

Q68. Solution
Correct Answer: (B)

The sentence with the same meaning is-In order to avoid getting late for office, I woke up early.

Q69. Solution
Correct Answer: (A)

The correct sequence of the jumbled sentences is DABC. Let’s analyze the logical flow: D introduces the main idea: "Prioritising drinking water source sustainability is one of the essential priorities identified." This sets the stage for discussing the importance of water source sustainability. A explains a limitation: "Unfortunately, it is narrowly defined as protecting water sources in the immediate vicinity." This highlights a problem: sustainability efforts are limited in scope. B expands on the issue: "The larger dimensions are often left out."This emphasizes that broader aspects of sustainability are ignored. C provides a solution: "Source sustainability also needs to be made a convergence priority." This suggests that a more integrated approach is required. Thus, the correct sequence DABC ensures a smooth transition from the main idea (D) → problem (A & B) → solution (C).

Q70. Solution
Correct Answer: (A)

- A introduces the central theme: "Green managements are about the environment and sustainability." This serves as the topic sentence, establishing the context of green management. - C explains the impact of green management: "It can cause enormous changes in companies’ operations to green their products, operations, and facilities." This sentence builds on the idea of how companies adapt to green practices. - D further elaborates on the benefits: "It can change their sustainability business to competitive advantages." This logically follows, explaining how green management can create a competitive edge. - B concludes by summarizing the key benefit: "Adopting green innovation by companies and organisations keeps the environment healthy and clean." This ties everything together, emphasizing the positive environmental impact.

Q71. Solution
Correct Answer: (C)

All except 'Pigeon' are flightless birds. Whereas 'Pigeon' is a flying bird.

Q72. Solution

Correct Answer: (A)

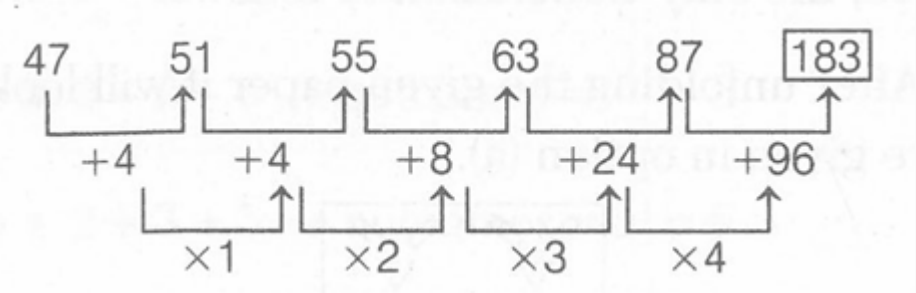
The figure that will replace the question mark (?) in the following figure series is shown below:

The series is rotating in clockwise direction. So, the next figure in the series will be option 1. Hence, 'option 1' is the correct answer.

Q73. Solution

Correct Answer: (B)

The pattern of the series is as shown below.



$\therefore \quad ? = 183$

Q74. Solution

Correct Answer: (C)

As, $19 : 12 \Rightarrow (12 \times 2 - 5) = 19$ Similarly, 23:? $\Rightarrow \quad (x \times 2 - 5) = 23$

$\Rightarrow \quad 2x - 5 = 23$

$\Rightarrow \quad 2x = 23 + 5 \quad \Rightarrow \quad x = \frac{28}{2} = 14$

Q75. Solution

Correct Answer: (B)

As, 'Bat' is a 'Mammal'. Similarly, 'Spiders' are 'Arthropoda'.

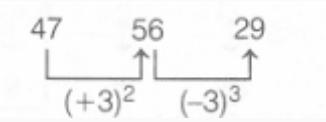
Q76. Solution

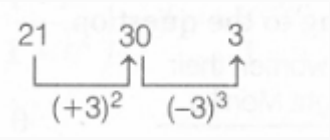
Correct Answer: (B)

As, the group of 'Grapes' is called 'Bunch'. Similarly, the group of 'Sheep' is called 'Herd'.

Q77. Solution

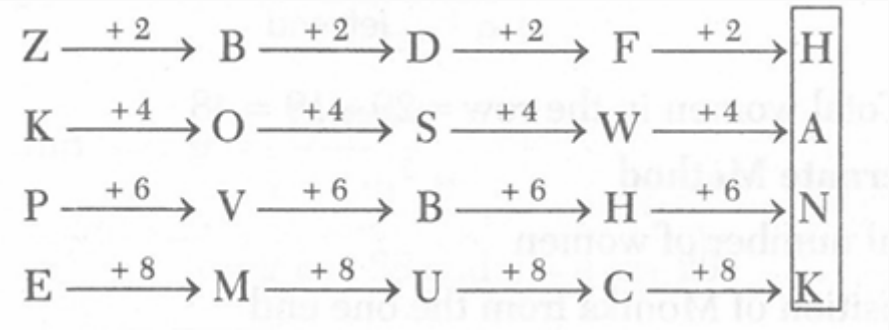
Correct Answer: (D)

As, 

Similarly, 

Q78. Solution

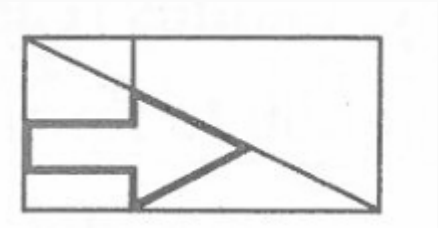
Correct Answer: (A)

Pattern of the series is as follows. 

$\therefore ? =$ HANK

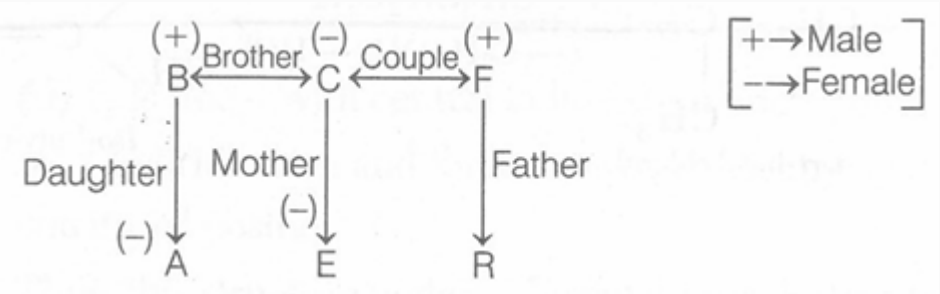
Q79. Solution

Correct Answer: (B)

The given question figure is completely embedded in the option figure (2). 

Q80. Solution

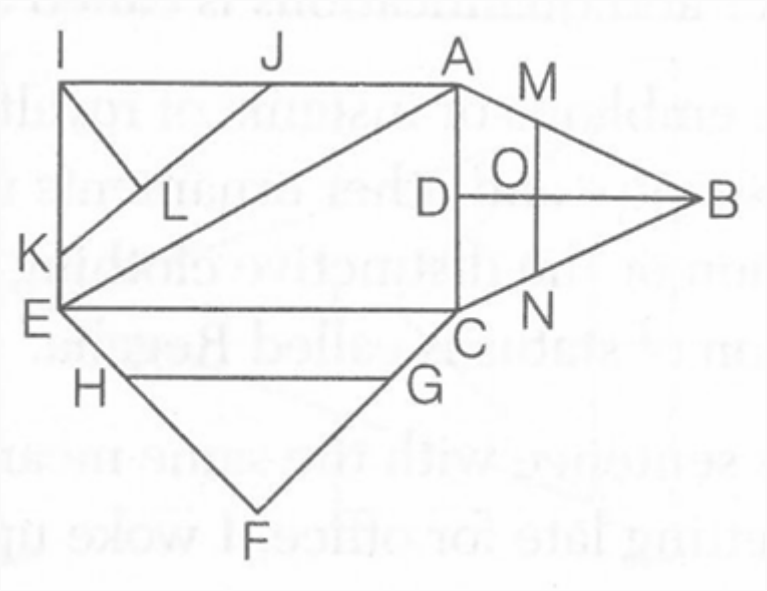
Correct Answer: (B)

According to the question, 

Clearly, ' A ' is the cousin of ' E ' is the correct statement.

Q81. Solution

Correct Answer: (D)



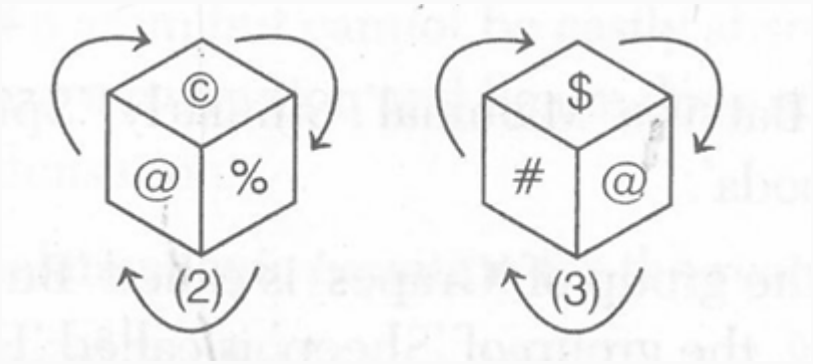
Naming of the figure,

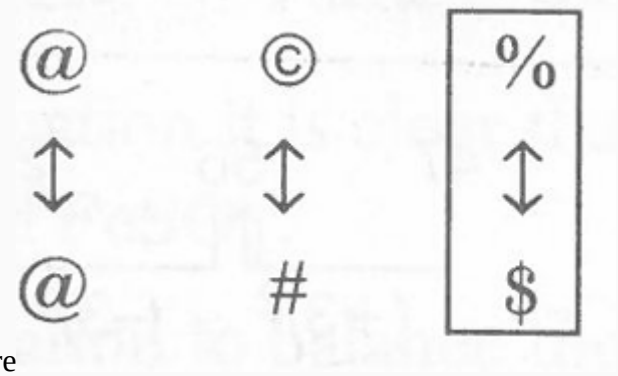
There are total 13 triangles : △KIL, △ILJ, △KIJ, △EIA, △EAC, △ADB, △DBC, △ABC, △HFG, △ECF, △BMN, △MOB and △ONB.

Q82. Solution

Correct Answer: (D)

From Dice (1) and (2) '%' and '@' is'common in both the dice, so '#' is opposite of '(0).. Now, from Dice (2) and (3) common symbol are '@â€™ and rotate it in clockwise direction from the common face.



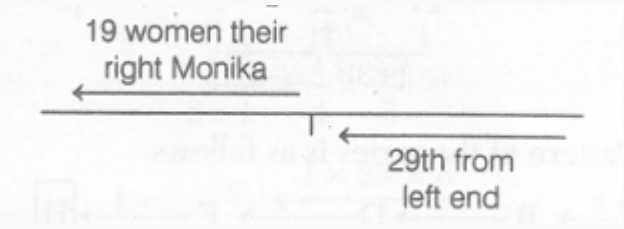


Opposite faces are

Hence, $ is at the bottom when ' % ' at the top.

Q83. Solution

Correct Answer: (B)



According to the question,

$\Rightarrow$ Total women in the row $= 29 + 19 = 48$

Alternate Method

Total number of women

$=$ Position of Monika from the one end

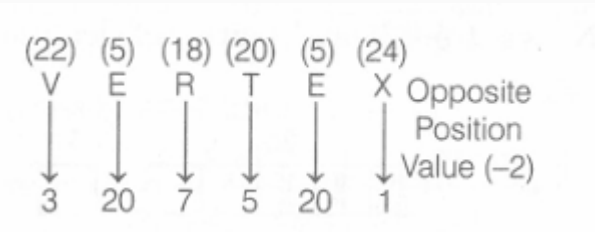
$+$ Number of women on her right side

$= 29 + 19 = 48$

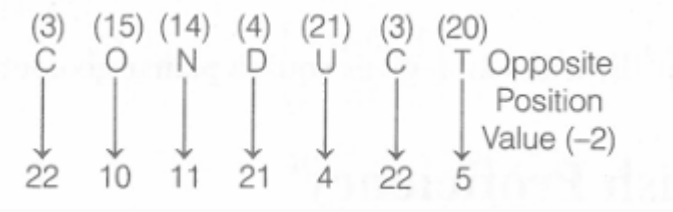
Q84. Solution

Correct Answer: (B)

According to the question, As,

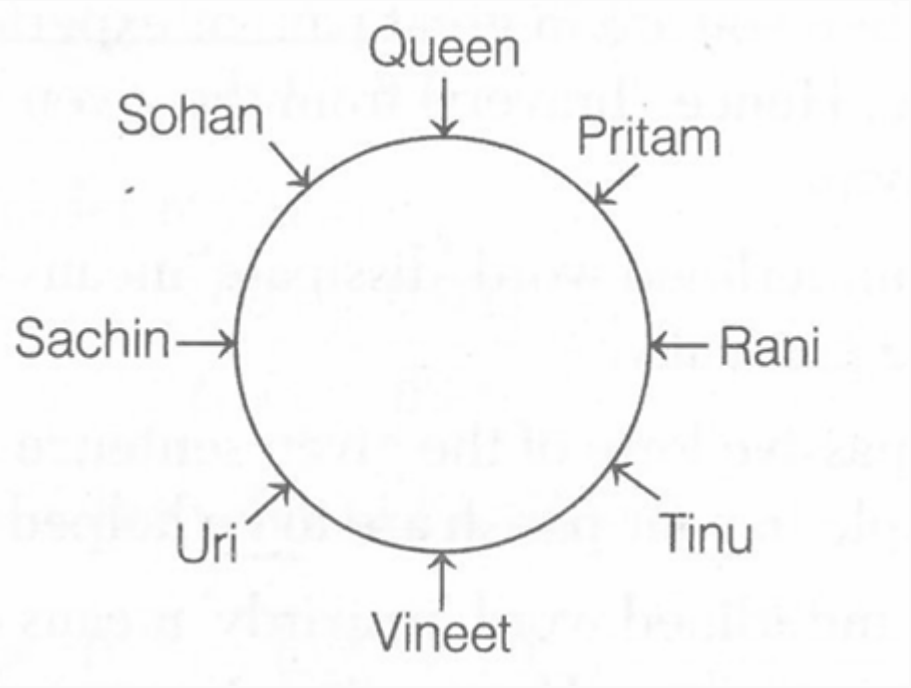


Similarly,



Q85. Solution

Correct Answer: (A)



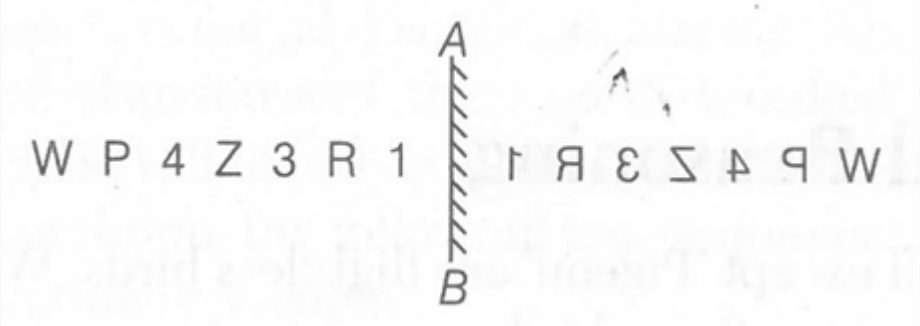
According to the question,

Hence, 'Tinu' is sitting immediate right of 'Vineet'.

Q86. Solution

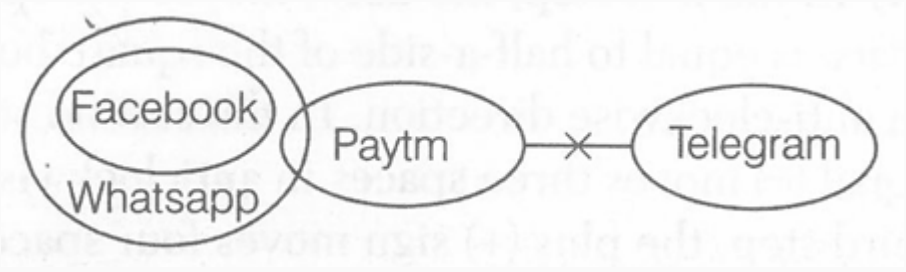
Correct Answer: (C)

The correct mirror image of the given question figure is as in option figure (c).



Q87. Solution

Correct Answer: (B)

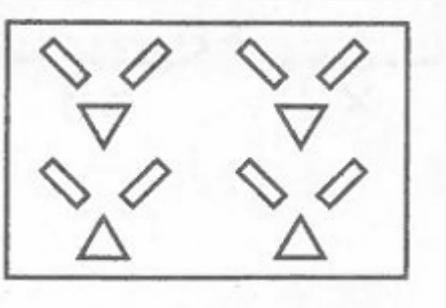


According to the statement,

Conclusions I. × II. ✓ Hence, the only Conclusion II follows.

Q88. Solution

Correct Answer: (A)



After unfolding the given paper it will look like figure given in option (1).

Q89. Solution

Correct Answer: (D)

As, nothing is mentioned about persnol likes or dislike of the manager, so conclusion I doesn't say anything about popularity of Satish. Hence, neither conclusion I nor conclusion II folllows.

Q90. Solution

Correct Answer: (A)

Let Anil's present age be x years. Then, Bishnu's present age $= \frac{240}{x} y$yr. According to question,

$$2\left(\frac{240}{x}\right) - x = 4$$

$$\Rightarrow \quad 480 - x^2 = 4x$$

$$\Rightarrow \quad x^2 + 4x - 480 = 0$$

$$\Rightarrow \quad (x+24)(x-20) = 0$$

$$\Rightarrow \quad x = 20$$

$\therefore$ Bishnu's present age $= \dfrac{240}{20}$ $\therefore$ Thus, Bishnu's age 2 yr ago $= 12$ years

$= 12 - 2 = 10$yr.

Q91. Solution

Correct Answer: (B)

$$\text{Given,} \quad \sin 9\theta = \sin\theta$$
$$\Rightarrow \quad \sin 9\theta - \sin\theta = 0$$
$$\Rightarrow \quad 2\cos\left(\frac{9\theta+\theta}{2}\right)\sin\left(\frac{9\theta-\theta}{2}\right) = 0$$
$$\left[\because \sin C - \sin D = 2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)\right]$$
$$\Rightarrow \quad 2\cos 5\theta \sin 4\theta = 0$$
$$\Rightarrow \text{ Either } \cos 5\theta = 0 \text{ or } \sin 4\theta = 0, \text{ also } \theta \in [0, 2\pi]$$
$$\Rightarrow \quad 5\theta = (2n+1)\frac{\pi}{2}$$
$$\text{or} \quad 4\theta = n\pi$$
$$\Rightarrow \quad \theta = (2n+1)\frac{\pi}{10} \text{ or } \theta = \frac{n\pi}{4}$$
$$\therefore \quad \theta = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{\pi}{2}, \frac{7\pi}{10}, \frac{9\pi}{10}, \frac{11\pi}{10}, \frac{13\pi}{10}, \frac{3\pi}{2}, \frac{17\pi}{10}, \frac{19\pi}{10}$$
$$\text{or} \quad 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}, 2\pi$$
$$\therefore \text{ Total number of solutions } = 17$$
$$\left(\because \frac{\pi}{2} \text{ and } \frac{3\pi}{2} \text{ are common}\right)$$

Q92. Solution

Correct Answer: (B)

Total outcomes $= 216$ Now, number of cases of getting 10 from 3 dices in single throw are Case $1: \ 1+3+6 \rightarrow$ outcomes $= 3! = 6$ Case $2: \ 1+4+5 \rightarrow$ outcomes $= 3\,! = 6$ Case $3: \ 2+2+6 \rightarrow$ outcomes $= \frac{3!}{2!} = 3$ Case $4: \ 2+3+5 \rightarrow$ outcomes $= 3! = 6$ Case $5: \ 2+4+4 \rightarrow$ outcomes $= \frac{3!}{2!} = 3$ Case $6: \ 3+3+4 \rightarrow$ outcomes $= \frac{3!}{2!} = 3$ Favourable outcomes $= 27 \therefore$ Probability $= \frac{27}{216} = \frac{1}{8}$

Q93. Solution

Correct Answer: (C)

We know that, if roots are of same sign $\Rightarrow$ Product of root is + ve if roots are of opposite sign $\Rightarrow$ Product of roots is -ve Here, $x^2 + bx - c = 0$ (b, $c > 0$) Product of roots $= \frac{-c}{1} = -$ve $\therefore$ Roots are of opposite sign.

Q94. Solution

Correct Answer: (D)

We know that AM $\geq$ GM...(i) Now, applying Eq. (i) in a and $b \Rightarrow \quad \frac{a+b}{2} \geq \sqrt{ab}$....(ii) applying Eq. (i) in b and $c \Rightarrow \quad \frac{b+c}{2} \geq \sqrt{bc}$....(iii) and applying Eq. (i) in c and $a \Rightarrow \quad \frac{c+a}{2} \geq \sqrt{ca}$....(iv) By E'qs. (ii), (iii) and (iv)

$$\Rightarrow \quad \frac{(a+b)(b+c)(c+a)}{8} \geq abc$$
$$\Rightarrow \quad (a+b)(b+c)(c+a) \geq 8abc$$

Q95. Solution

Correct Answer: (A)

Given $x = a\left(\frac{1-t^2}{1+t^2}\right)$ and $y = \frac{2at}{1+t^2}$

$$\text{Let} \quad t = \tan\theta$$
$$\Rightarrow \quad x = a\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)$$
$$\text{and} \quad y = \frac{2a\tan\theta}{1+\tan^2\theta}$$
$$\Rightarrow \quad x = a\cos 2\theta \text{ and } y = a\sin 2\theta$$
$$\Rightarrow \cos 2\theta = \frac{x}{a} \text{ and } \sin 2\theta = \frac{y}{a}$$

$\cos^2 2\theta = \frac{x^2}{a^2}$....(i) and $\sin^2 2\theta = \frac{y^2}{a^2}$....(ii) Adding Eqs. (i) and (ii), we get

Squaring both sides, we get

$$\cos^2 2\theta + \sin^2 2\theta = \frac{x^2}{a^2} + \frac{y^2}{a^2} \therefore$$
$$\Rightarrow \quad x^2 + y^2 = a^2$$

Locus is a circle having centre at origin and radius a.

Q96. Solution

Correct Answer: (A)

$$\mathrm{A} + \mathrm{B} + \mathrm{C} = 180^\circ \text{ (Angle sum property of a triangle)}$$
$$\Rightarrow \mathrm{A} + \mathrm{B} = 180^\circ - \mathrm{C}$$
$$\Rightarrow \frac{A+B}{2} = 90^\circ - \frac{C}{2}$$
$$\Rightarrow \tan\left(\frac{A}{2} + \frac{B}{2}\right) = \tan\left(90^\circ - \frac{C}{2}\right)$$
$$\Rightarrow \frac{\tan A/2 + \tan B/2}{1 - \tan A/2 \tan B/2} = \cot C/2$$
$$\Rightarrow \frac{\frac{1}{3} + \frac{2}{3}}{1 - \frac{1}{3} \times \frac{2}{3}} = \cot C/2 \Rightarrow \frac{1}{1 - \frac{2}{9}} = \cot C/2$$
$$\Rightarrow \frac{9}{7} = \cot C/2 \Rightarrow \tan C/2 = 7/9$$

Q97. Solution

Correct Answer: (A)

We know that, if A is an orthogonal matrix, then

$$\Rightarrow \quad AA^T = I$$

$$\Rightarrow \quad \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} 1 & 2 & a \\ 2 & 1 & 2 \\ 2 & -2 & b \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix} \begin{bmatrix} 1 & 2 & a \\ 2 & 1 & 2 \\ 2 & -2 & b \end{bmatrix} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

By

$$\begin{bmatrix} 9 & 0 & a+4+2b \\ 0+4+2b & 9 & 2a+2-2b \\ 2a+2-2b & a^2+4+b^2 & \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$\Rightarrow \quad a+4+2b = 0 \text{ and } 2a+2-2b = 0$$

$$a+2b = -4 \ldots (i)$$

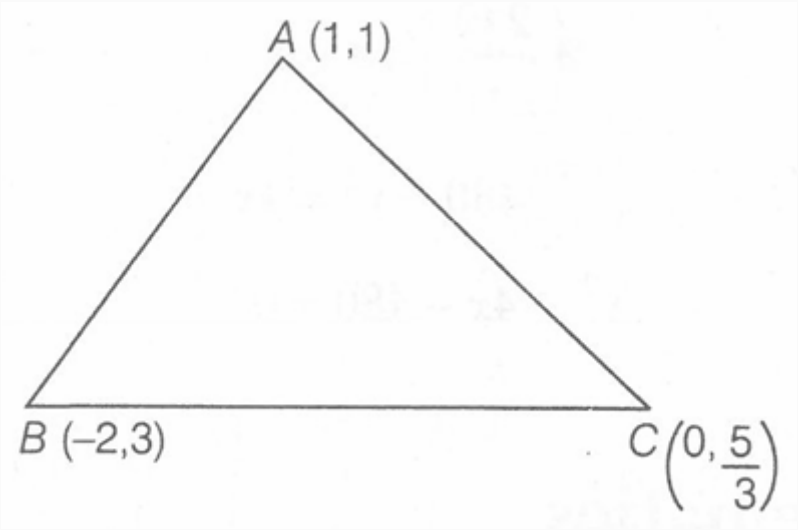
$$a-b = -1 \ldots (ii)$$

solving Eqs. (i) and (ii), we get $a = -2, b = -1$

Q98. Solution

Correct Answer: (C)

According to question,



Let find slope of AB, BC and CA Slope of $AB = \frac{1-3}{1-(-2)} = \frac{-2}{3}$ Slope of $BC = \frac{3-\left(\frac{5}{3}\right)}{-2-0} = \frac{-2}{3}$ Slope of $CA = \frac{\frac{5}{3}-1}{0-1} = \frac{-2}{3}$ Since, slope of all lines are same, therefore, points are collinear.

Q99. Solution

Correct Answer: (D)

Here, $[abc] = |a \cdot (b \times c)|$ Now, according to question,

$$[a \quad bc] = \begin{matrix} 1 & 0 & -1 \\ x & 1 & 1-x \\ y & x & 1+x-y \end{matrix}$$

$$\begin{aligned} &= 1(1+x-y-x(1-x)) - 1\left(x^2-y\right) \\ &= 1+x-y-x+x^2-x^2+y \\ &= 1 \\ &= \text{ Independent of } x \text{ and } y. \end{aligned}$$

Q100. Solution

Correct Answer: (A)

We know that, projection of $\mathbf{a}$ on $\mathbf{b}$ is given by projection, $|\mathrm{a}| \cos\theta = \frac{(\mathbf{a} \cdot \mathbf{b})}{|\mathbf{b}|}$ Let line joining points $(3,4,5)$ and $(4,6,3)$ is L_1 and line joining points $(-1,2,4)$ and $(1,0,5)$ is $L_2 \Rightarrow$ $\begin{matrix} L_1 = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}} \\ L_2 = 2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}} \end{matrix}$ $\therefore$ Projection of L_1 and $L_2 = \frac{L_1 \cdot L_2}{|L_2|} = \frac{2-4-2}{\sqrt{4+4+1}} = \frac{-4}{3}$ $\therefore$ Magnitude $= \frac{4}{3}$

Q101. Solution

Correct Answer: (D)

$$\begin{aligned} &\text{Given, } \sim (p \vee q) \vee (\sim p \wedge q) \\ &\Rightarrow (\sim p \wedge \sim q) \vee (\sim p \wedge q) \text{ (By De Morgan's Law)} \\ &\Rightarrow \sim p \wedge (\sim q \vee q) \text{ (Distributive Law)} \\ &\Rightarrow \sim p \wedge t \text{ (By complement law, where } t = \text{ tautology)} \\ &\Rightarrow \sim p \text{ (Identity law)} \end{aligned}$$

Q102. Solution

Correct Answer: (B)

$$\begin{aligned} \text{Let } S &= \sum_{k=1}^{n} k(k+1)(k-1) \\ &= \sum_{k=1}^{n} k^3 - k \end{aligned}$$

We know that $\sum_{r=1}^{P} r = \frac{P(P+1)}{2}$ and $\sum_{r=1}^{P} r^3 = \left[\frac{P(P+1)}{2}\right]^2$

$$\begin{aligned} S &= \left[\frac{n(n+1)}{2}\right]^2 - \left[\frac{n(n+1)}{2}\right] \\ &= \frac{n(n+1)}{2}\left[\frac{n(n+1)}{2} - 1\right] \\ &= \left(\frac{n^2+n}{2}\right)\left[\frac{n^2+n-2}{2}\right] \\ &= \frac{n^4}{4} + \frac{n^3}{2} - \frac{n^2}{4} - \frac{n}{2} \ldots (i) \end{aligned}$$

Now, by comparing $pn^4 + qn^3 + tn^2 + sn$ with Eq. (i), $S = -\frac{1}{2}$

Q103. Solution

Correct Answer: (C)

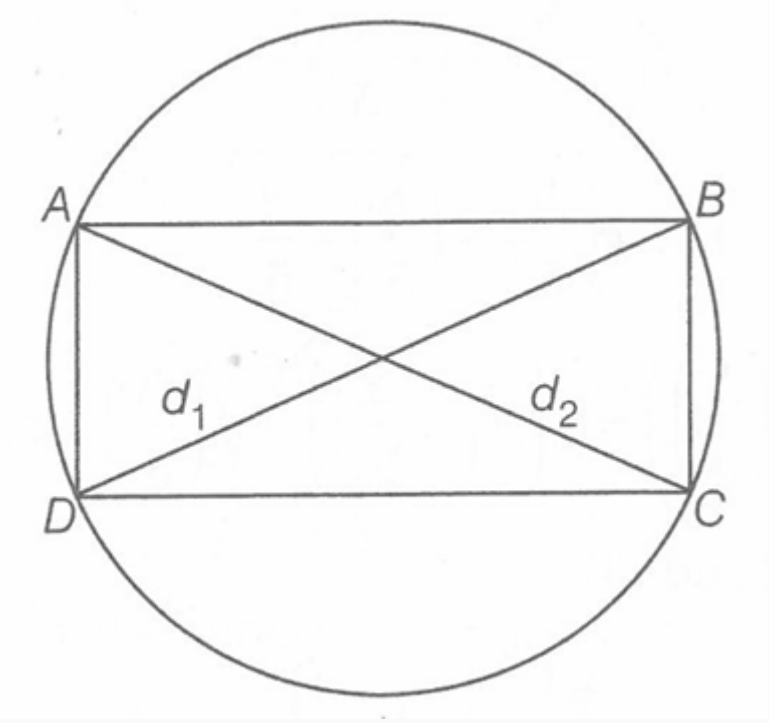
$$\begin{aligned} \text{Mean } &= np = 4 \\ \text{Variance } &= npq = 3 \end{aligned}$$ and $p = 1 - q = 1 - \frac{3}{4} = \frac{1}{4} \Rightarrow \quad n = 16$ Now, Mode of Binomial distribution is given by $(n+1)P$ Case IIf $(n+1)P = \text{Integer}(I)$, then Mode $= \{I, I-1\}$ Case II If $(n+1)P \neq \text{Integer}(I+f)$, then Mode $= \{I\}$

$$\Rightarrow \quad q = \frac{3}{4}$$

$$\begin{aligned} \therefore (n+1)P &= (16+1)\frac{1}{4} \\ &= \frac{17}{4} = 4.25 \\ &= 4 + 0.25 \\ \Rightarrow \text{ Mode } &= 4 \end{aligned}$$

Q104. Solution

Correct Answer: (B)



According to question,

Diameter $= R$ Diagonals, $d_1 = d_2 = R$

$$\begin{aligned} \text{Max Area of rectangle (any quadrilateral) } &= \frac{1}{2} d_1 \times d_2 \\ &= \frac{1}{2} \times R \times R = \frac{R^2}{2} \end{aligned}$$

Q105. Solution

Correct Answer: (B)

$$\begin{aligned} \text{Given, } n(R) &= 794 \\ n(T) &= 187 \\ n(R \cup T)' &= 63 \\ n(\text{Total}) &= n(R \cup T) + n(R \cup T)' \\ \Rightarrow \quad 1003 &= n(R \cup T) + 63 \\ \Rightarrow \quad n(R \cup T) &= 940 \end{aligned}$$

By set theory

$$\begin{aligned} \Rightarrow \quad n(R \cup T) &= n(R) + n(T) - n(R \cap T) \\ \Rightarrow \quad 940 &= 794 + 187 - n(R \cap T) \\ \Rightarrow \quad n(R \cap T) &= 981 - 940 = 41 \end{aligned}$$

Q106. Solution

Correct Answer: (B)

Given, Hyperbola $= 4x^2 - 9y^2 - 1 = 0$ $\Rightarrow \dfrac{x^2}{\left(\frac{1}{4}\right)} - \dfrac{y^2}{\left(\frac{1}{9}\right)} = 1$ $\Rightarrow \dfrac{x^2}{\frac{1}{2}^2} - \dfrac{y^2}{\frac{1}{3}^2} = 1$ We know that. if hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

then Eccentricity: $e = \sqrt{1 + \frac{b^2}{a^2}}$ and focus $\equiv (\pm \text{be}, 0) = \left(\pm \dfrac{1}{2} \times \dfrac{\sqrt{1/3}}{3}, 0\right) = \left(\pm \dfrac{\sqrt{13}}{6}, 0\right)$

Q107. Solution

Correct Answer: (D)

Given, $f(x) = [x]$ and $g(x) = |x|$ Now, $f\left(\dfrac{-8}{5}\right) = \left[-\dfrac{8}{5}\right] = -2 \ldots (i)$ $g\left(\dfrac{8}{5}\right) = \dfrac{8}{5} = \dfrac{8}{5} \ldots (ii)$ Now,

$f\left(g\left(\frac{8}{5}\right)\right) - g\left(f\left(-\frac{8}{5}\right)\right) = f\left(\frac{8}{5}\right) - g(-2)$ By Eqs. (i) and (ii), $= \left[\dfrac{8}{5}\right] - |-2|$ $= 1 - 2$ $= -1$

Q108. Solution

Correct Answer: (D)

$f(x) = \begin{cases} \frac{x^2-1}{x^2-2x+1}; & x > 1 \\ \frac{1}{2} \quad ; & x = 1 \\ \frac{x^2-1}{x^2+2x-3}; & x < 1 \end{cases}$ $\because \quad x^2 + 3x - x - 3 = x(x+3) - 1(x+3)$ Lets Find LHL, $\lim_{x \to 1^-} \frac{x^2-1}{x^2-2x+1}$

and RHL, $\lim_{x \to 1^+} \frac{x^2-1}{x^2+2x-3}$ $\Rightarrow \lim\limits_{x \to 1^-} \dfrac{(x-1)(x+1)}{(x-1)^2} \Rightarrow \lim\limits_{x \to 1^+} \dfrac{(x-1)(x+1)}{(x+3)(x-1)}$ $\Rightarrow \lim\limits_{x \to 1^-} \dfrac{x+1}{x-1} = \infty \Rightarrow \lim\limits_{x \to 1^+} \dfrac{x+1}{x+3} = \dfrac{2}{4} = \dfrac{1}{2}$ $\therefore$ LHL $\neq f(1) \Rightarrow f(x)$ is not continuous at $x = 1$

Q109. Solution

Correct Answer: (C)

Given, $x\sqrt{1+y} + y\sqrt{1+x} = 0$

$\Rightarrow \quad x\sqrt{1+y} = -y\sqrt{1+x}$

$\Rightarrow \quad (x\sqrt{1+y})^2 = (-y\sqrt{1+x})^2$

$\Rightarrow \quad x^2(1+y) = y^2(1+x)$

$\Rightarrow \quad x^2 - y^2 = y^2x - x^2y$

$\Rightarrow \quad (x-y)(x+y) = xy(y-x)$

$\Rightarrow \quad x + y = -xy$

$\Rightarrow \quad x + y + xy = 0$

$\Rightarrow \quad y = \frac{-x}{1+x} = -1 + \frac{1}{1+x}$

Differentiating w.r.t, x $\frac{dy}{dx} = -\frac{1}{(1+x)^2}$

Q110. Solution

Correct Answer: (D)

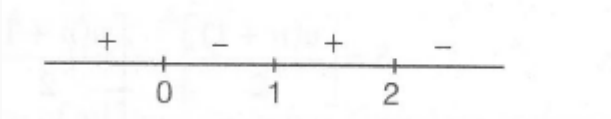
Given, $f(x)=\begin{cases}\dfrac{x-1}{x^2}; x\geq 1\\ \dfrac{-x+1}{x^2}; x<1\end{cases}$

$$f(x)=\begin{cases}\frac{1}{x}-\frac{1}{x^2}; & x\geq 1\\ -\frac{1}{x}+\frac{1}{x^2}; & x<1\end{cases}$$

$$f'(x)=\begin{cases}\frac{-1}{x^2}+\frac{2}{x^3}; & x\geq 1\\ \frac{1}{x^2}-\frac{2}{x^3}; & x<1\end{cases}$$

By observation, $f'(x)$ will be

$$f'(x)=\begin{cases}\frac{2-x}{x^3}; & x\geq 1\\ \frac{x-2}{x^3}; & x<1\end{cases}$$



Decreasing in $(0,1)\cup(2,\infty)$ and increasing in $(-\infty,0)\cup(1,2)$

Q111. Solution

Correct Answer: (B)

Let 3 numbers in AP are $a,(a+6),(a+12)$ Also, first and last number out of 4 numbers are equal $\therefore$ 4 numbers are $(a+12),a,(a+6),(a+12)$ Also, given, first 3 numbers are in GP

$$\Rightarrow \quad a^2=(a+12)(a+6)$$
$$\Rightarrow \quad a^2=a^2+18a+72$$
$$\Rightarrow \quad a=-\tfrac{72}{18}=-4$$

$\therefore$ 4 numbers are $8,-4,2,8$

Q112. Solution

Correct Answer: (C)

$$I=\int_0^\infty \frac{1}{(x^2+a^2)(x^2+b^2)}dx$$

$$I=\frac{1}{a^2-b^2}\int_0^\infty \frac{(x^2+a^2)-(x^2+b^2)}{(x^2+a^2)(x^2+b^2)}dx$$

$$I=\frac{1}{a^2-b^2}\int_0^\infty \frac{1}{(x^2+b^2)}-\frac{1}{(x^2+a^2)}dx$$

Let $I=\dfrac{1}{a^2-b^2}\left[\dfrac{1}{b}\tan^{-1}\dfrac{x}{b}-\dfrac{1}{a}\tan^{-1}\dfrac{x}{a}\right]_0^\infty$

$$I=\frac{1}{a^2-b^2}\left[\frac{1}{b}\times\frac{\pi}{2}-\frac{1}{a}\times\frac{\pi}{2}\right]$$

$$I=\frac{1}{(a+b)(a-b)}\left[\frac{a-b}{ab}\right]\times\frac{\pi}{2}$$

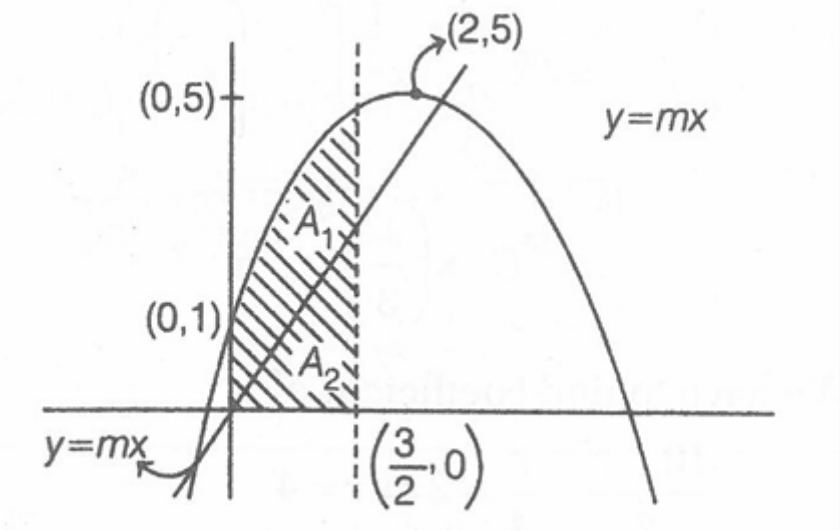
$$I=\frac{\pi}{2ab(a+b)}$$

Q113. Solution

Correct Answer: (A)

Given, $y = 1 + 4x - x^2 \Rightarrow \frac{dy}{dx} = 4 - 2x = 0 \Rightarrow x = 2[\text{point of maxima}]$ when, $x = 0, y = 1 \therefore$ Graph

$\Rightarrow \quad y_{\max} = 1 + 4 \times 2 - 4 = 5$



according to question and above information

$\Rightarrow$ If $y = mx$ bisects the area bounded

$\Rightarrow \quad A_1 = A_2$

$$\Rightarrow \quad \int_0^{3/2} (1 + 4x - x^2)dx = 2\int_0^{3/2} mxdx$$

$$\Rightarrow \quad \left[x + 2x^2 - \frac{x^3}{3}\right]_0^{3/2} = \left[mx^2\right]_0^{3/2}$$

$$\Rightarrow \frac{3}{2} + 2 \times \frac{9}{4} - \frac{27}{8} \times \frac{1}{3} = m \times \frac{9}{4}$$

$$\therefore \quad m = \frac{13}{6}$$

Q114. Solution

Correct Answer: (C)

Given, $\frac{dy}{dx} - \frac{y}{x+1} = e^{3x}(x+1)$ Above is linear differential equation of form

$$\frac{dy}{dx} + Px = Q$$

$\Rightarrow$ Solution will be

$$\therefore \quad \text{IF} = e^{\int Pdx} = e^{\int -\frac{1}{x+1}dx} = e^{-\ln(1+x)} = \frac{1}{1+x}$$

$$y \cdot \text{IF} = \int Q \cdot \text{IF}dx$$

$$y \cdot \frac{1}{1+x} = \int e^{3x}(x+1) \times \frac{1}{(1+x)}dx$$

$$\frac{y}{1+x} = \frac{e^{3x}}{3} + C'$$

$$\frac{3y}{1+x} = e^{3x} + C' \qquad [\because C = 3C']$$

Q115. Solution

Correct Answer: (B)

$$\begin{aligned} \text{Let} \quad & \log_2|x| = t \ldots (i) \\ \therefore \quad & \sqrt{5-t} = 3-t \\ \Rightarrow \quad & 5-t = (3-t)^2 \\ \Rightarrow \quad & 5-t = 9+t^2-6t \end{aligned}$$

$$\begin{aligned} \Rightarrow \quad & t^2-5t+4=0 \\ \Rightarrow \quad & (t-4)(t-1)=0 \\ \Rightarrow \quad & t=1 \text{ or } 4[\text{rejected}] \end{aligned}$$

$[\because \sqrt{5-4} = 3-4, 1 \neq -1]$ $\therefore$ Total 2 real solutions.

$$\begin{aligned} \Rightarrow \quad & \log_2|x| = 1 \\ \Rightarrow \quad & |x| = 2 \\ \Rightarrow \quad & x = \pm 2 \end{aligned}$$

Q116. Solution

Correct Answer: (A)

Let $I = \int_0^{\frac{\pi}{2}} \log(\tan x)dx$....(i) By applying property,

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

$$I = \int_0^{\frac{\pi}{2}} \log\left(\tan\left(\frac{\pi}{2}-x\right)\right)dx \text{ Adding Eqs.}$$

$$I = \int_0^{\frac{\pi}{2}} \log(\cot x)dx \ldots (ii)$$

(i) and (ii),

$$2I = \int_0^{\frac{\pi}{2}} [\log(\tan x) + \log(\cot x)]dx$$

$$2I = \int_0^{\frac{\pi}{2}} \log 1 dx = 0$$

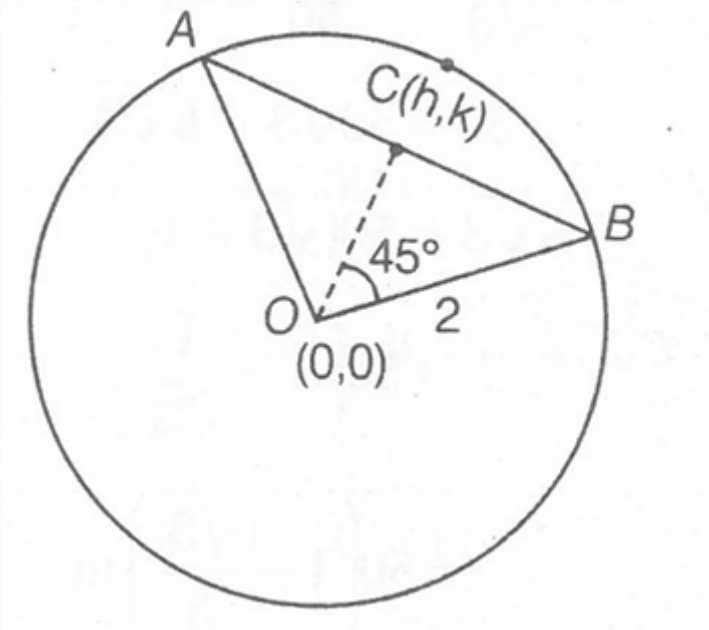
Q117. Solution

Correct Answer: (B)

To optimised n number of parameters, we need atleast n constraints.

Q118. Solution

Correct Answer: (C)



According to question,

$$\cos 45^\circ = \frac{\sqrt{h^2+k^2}}{2}$$

$OC = \sqrt{h^2+k^2}$ In $\triangle OCB$, $\Rightarrow \quad \frac{\sqrt{h^2+k^2}}{2} = \frac{1}{\sqrt{2}}$ Replacing $h \to x$ and $k \to y$ Locus $\Rightarrow x^2+y^2=2$

$$\Rightarrow \quad h^2+k^2=2$$

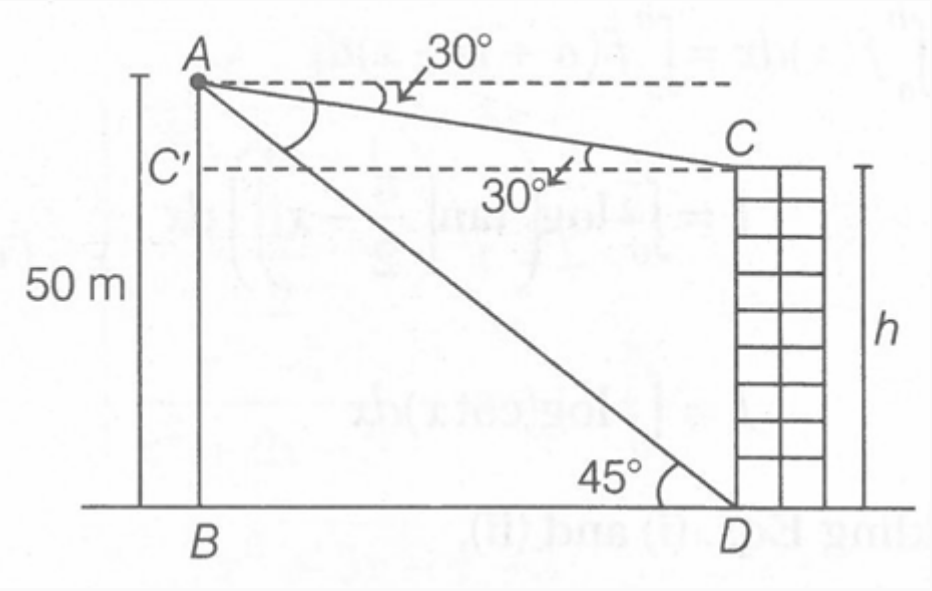
Q119. Solution

Correct Answer: (B)

Total person $= 10$ Two groups are formed containing 4 and 6 persons. $\therefore$ Total ways of formation of groups $= \frac{10!}{4!6!}$ Now, 4 persons can be arranged on round table $= (4-1)! = 3!$ and 6 persons can be arranged on round table $= (6-1)\,! = 5!$ $\therefore$ Total ways of arranging the guest $= \frac{10!}{4!6!} \times 3! \times 5! = \frac{10!}{24}$

Q120. Solution

Correct Answer: (D)



Accoding to condition,

Let, height of tower $= h$ In $\triangle ABD$, $\tan 45^\circ = \frac{AB}{BD} \Rightarrow BD = 50$ m Now, In $\triangle ACC'$

$$\tan 30^\circ = \frac{AC'}{C'C}$$

$$\Rightarrow \quad \frac{1}{\sqrt{3}} = \frac{50-h}{50}$$

$$\Rightarrow \quad 50 = 50\sqrt{3} - h\sqrt{3}$$

$$\Rightarrow \quad h\sqrt{3} = 50(\sqrt{3}-1)$$

$$h = 50\left(1 - \frac{1}{\sqrt{3}}\right)$$

$$= 50\left(1 - \frac{\sqrt{3}}{3}\right)\text{m}$$

Q121. Solution

Correct Answer: (D)

$$y = \tan^{-1}\left(\frac{\sqrt{x}-x}{1+x^{\frac{3}{2}}}\right)$$

Given, $y = \tan^{-1}\left(\frac{\sqrt{x}-x}{1+\sqrt{x}\cdot x}\right)$ Differentiating w.r.t x

$$y = \tan^{-1}(\sqrt{x}) - \tan^{-1} x$$

$$\left[\because \tan^{-1}\left(\frac{x-y}{1+xy}\right) = \tan^{-1} x - \tan^{-1} y\right]$$

$y'(x) = \frac{dy}{dx} = \frac{1}{1+x} \times \frac{1}{2\sqrt{x}} - \frac{1}{1+x^2}$ Now, $y'(1) = \frac{1}{2} \times \frac{1}{2} - \frac{1}{2} = -\frac{1}{4}$

Q122. Solution

Correct Answer: (A)

General term of given binomial expansion is $T_{r+1} = {}^{10}C_r \left(\frac{1}{3} x^{\frac{1}{2}}\right)^{10-r} \left(x^{-\frac{1}{4}}\right)^r$ $= {}^{10}C_r \times \left(\frac{1}{3}\right)^{10-r} x^{\frac{10-r}{2} - \frac{r}{4}}$ We have to find coefficient of x^2

$$\frac{10-r}{2} - \frac{r}{4} = 2 \Rightarrow r = 4$$

$$\Rightarrow \quad T_{4+1} = {}^{10}C_4 \left(\frac{1}{3}\right)^6 x^2$$

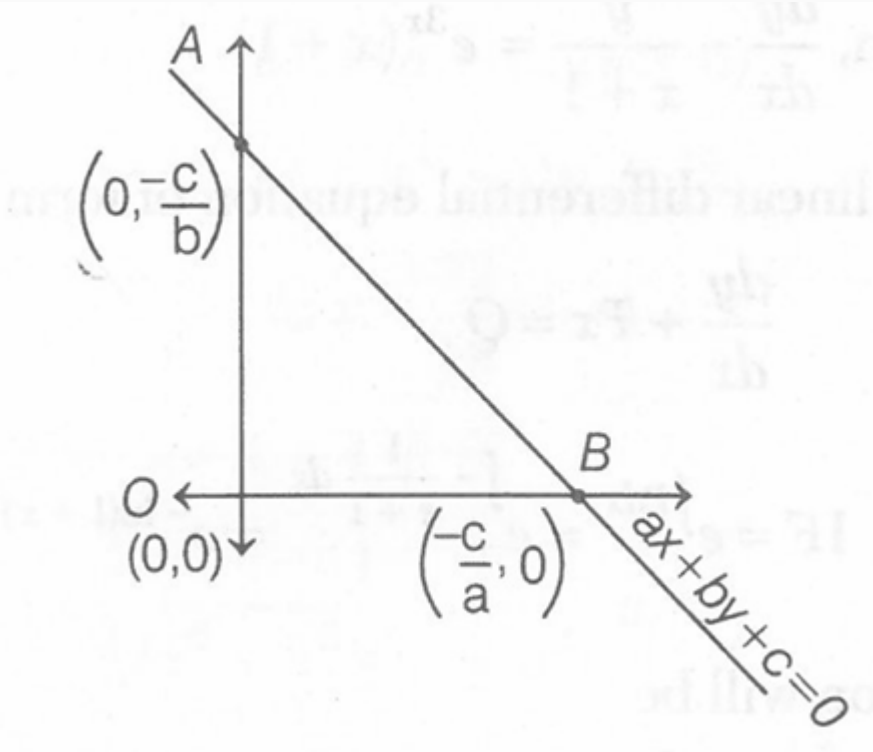
$$\therefore \text{ Coefficient of } x^2 = {}^{10}C_4 \left(\frac{1}{3}\right)^6$$

$$= \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} \times \frac{1}{3^6} = \frac{70}{243}$$

Q123. Solution

Correct Answer: (C)

Given a, c, b are in GP $\Rightarrow \quad c^2 = ab$...(i) According to question,



$$\therefore \text{ Area of } \triangle OAB = \frac{1}{2} \times OA \times OB$$

$$= \frac{1}{2} \times \left(\frac{-c}{b}\right) \times \left(\frac{-c}{a}\right)$$

$$= \frac{1}{2} \times \frac{c^2}{ab} = \frac{1}{2} [by Eq. (i)]$$

Q124. Solution

Correct Answer: (C)

Here we have 4 odd digits $(3,3,5,5)$ and 5 even digits (2, 2, 8, 8, 8). $\bar{O}\bar{E}\bar{O}\bar{E}\bar{O}\bar{O}\bar{E}\bar{O}$ where, $E \rightarrow$ even place and $O \rightarrow$ odd place $\Rightarrow$ Number of ways odd digits will be place on even places $= \frac{4!}{2!2!}$ Number of ways even digits will be place on odd places $= \dfrac{5!}{2!3!}$

$$\therefore \text{ Total number of ways } = \frac{4!}{2!2!} \times \frac{5!}{2!3!} = 60$$

Q125. Solution

Correct Answer: (C)

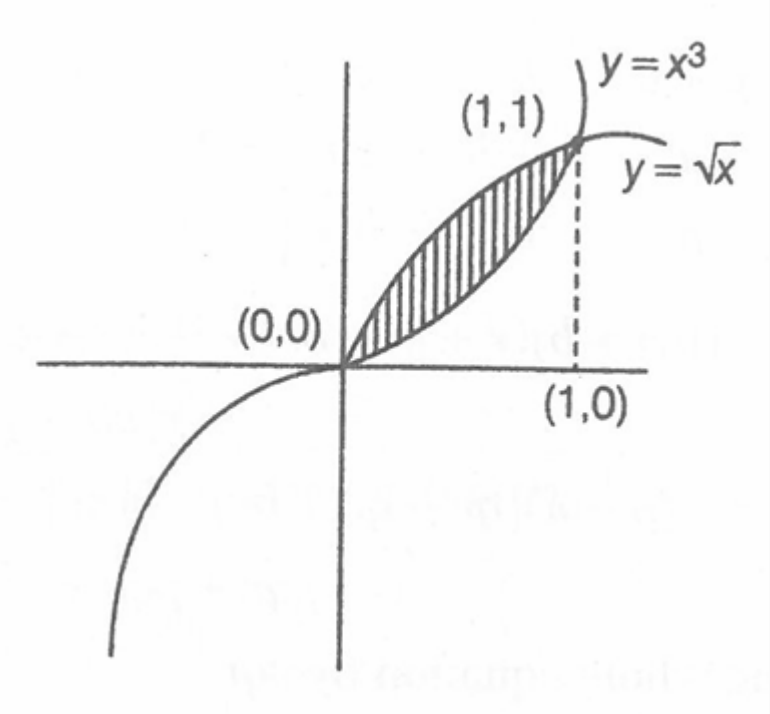
Given, $A = \begin{bmatrix} 3 & -2 & 4 \\ 1 & 2 & -1 \\ 0 & 1 & 1 \end{bmatrix}$ and $A^{-1} = \frac{1}{k}\text{adj}(A)$....(i) We know that, $A^{-1} = \frac{\text{adj}(A)}{|A|}$...(ii) By comparing Eqs. (i) and (iii), $k = |A|$ So, $|A| = \begin{matrix} 3 & -2 & 4 \\ 1 & 2 & -1 \\ 0 & 1 & 1 \end{matrix}$

$$= 3(2+1) + 2(1+0) + 4(1-0) = 15$$

Q126. Solution

Correct Answer: (C)

Given curves are $y = x^3$ and $y = \sqrt{x}$ Both curves intersects at $x = 0$ and $x = 1$



$$A = \int_0^1 \left(x^3 - \sqrt{x}\right) dx = \left[\frac{x^4}{4} - \frac{2x^{3/2}}{3}\right]_0^1$$

Shaded region is required area $= \left[\dfrac{1}{4} - \dfrac{2}{3}\right]$

$$= \frac{5}{12} \text{ sq. unit}$$

Q127. Solution

Correct Answer: (A)

Let $|A| = \begin{vmatrix} p+a & b & c \\ a & q+b & c \\ a & b & r+c \end{vmatrix}$ Also, given that equations has non trivial solution $|A| = 0$

$$\Rightarrow \begin{vmatrix} p+a & b & c \\ a & q+b & c \\ a & b & r+c \end{vmatrix} = 0$$

$\Rightarrow (p+a)[(q+b)(r+c) - bc] - b[a(r+c) - ca]$ Dividing whole equation by pqr

$+ c[ab - a(q+b)] = 0$

$\Rightarrow \quad (p+a)[qr + qc + br] - b[ar] + c[-aq] = 0$

$\Rightarrow \quad pqr + pqc + pbr + aqr = 0$

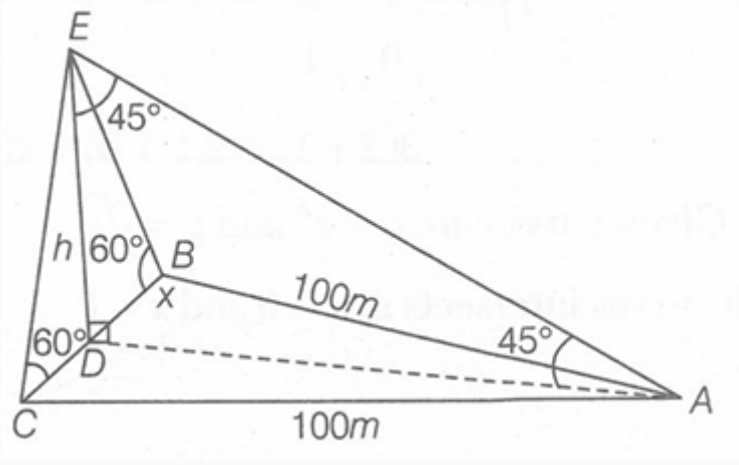
$$\Rightarrow \frac{pqr}{pqr} + \frac{pqc}{pqr} + \frac{prb}{pqr} + \frac{qra}{pqr} = 0$$

$$\Rightarrow 1 + \frac{c}{r} + \frac{b}{q} + \frac{a}{p} = 0$$

$$\Rightarrow \frac{a}{p} + \frac{b}{q} + \frac{c}{r} = -1$$

Q128. Solution

Correct Answer: (B)



According to question,

Let $DE = h$ and $CD = DB = x$ In $\triangle EBD$, $\tan 60^\circ = \frac{h}{x} \Rightarrow x = \frac{h}{\sqrt{3}}$ Now, in $\triangle ADE$ $\tan 45^\circ = \dfrac{ED}{DA}$

$\Rightarrow \quad DA = h$

In $\triangle ABD$, by applying Pythagoras theorem

$$\Rightarrow \left(\frac{h}{\sqrt{3}}\right)^2 + h^2 = 100^2$$

$$\Rightarrow \frac{4h^2}{3} = 10000 \quad \Rightarrow h = 50\sqrt{3}$$

Q129. Solution

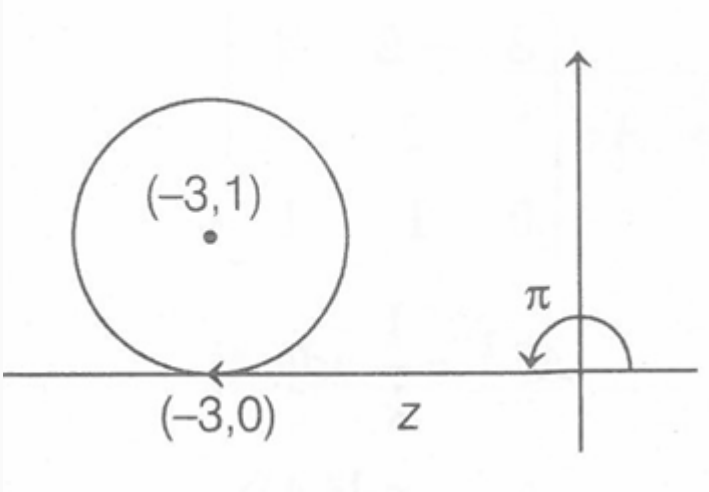
Correct Answer: (A)

$$\Rightarrow \quad |(x+3)+(y-1)i| = 1$$

Let $z = x + iy$ Given, $\quad |z+3-i| = 1 \Rightarrow \sqrt{(x+3)^2+(y-1)^2} = 1$ Above is the equation of circle

$$\Rightarrow \quad (x+3)^2+(y-1)^2 = 1 \ldots (i)$$

with centre $\equiv (-3, 1)$ and radius $= 1$ Also, $\arg(z) = \pi \Rightarrow \tan^{-1} \frac{y}{x} = \pi \Rightarrow \quad \frac{y}{x} = \tan\pi = 0 \Rightarrow y = 0$



$$\Rightarrow x = -3$$

By Eq. (i), $\therefore z = -3 + 0i$

$$\Rightarrow \text{Modulus of } z = 3$$

Q130. Solution

Correct Answer: (B)

Using $l_E = l_B + l_C$

$\Rightarrow l_B = l_E - l_C$

$= 6.6 \times 10^{-3} - 60 l_B$

$\Rightarrow \quad l_B + 60 l_B = 6.6 \times 10^{-3}$

or $\quad 61 l_B = 6.6 \times 10^{-3}$

$$\therefore \quad l_B = \frac{6.6 \times 10^{-3}}{61} \simeq 0.108 \text{ mA}$$

$= 0.108 \times 10^{-3}$

$= 0.108$ mA

Q131. Solution

Correct Answer: (C)

Given, $\frac{e^{7x}+e^x}{e^{3x}} \Rightarrow e^{4x} + e^{-2x}$ Series of $e^a = 1 + \frac{a}{1!} + \frac{a^2}{2!} + \frac{a^3}{3!} \ldots$

$$\Rightarrow e^{4x}+e^{-2x} = \left(1 + \frac{4x}{1!} + \frac{(4x)^2}{2!} + \frac{(4x)^3}{3!} + \ldots\right) + \left(1 + \frac{(-2x)}{1!} + \frac{(-2x)^2}{2!} + \ldots\ldots\right)$$

Here, coefficient of

$$x^2 \equiv \frac{4^2}{2!} + \frac{(-2)^2}{2!}$$

$$x^3 \equiv \frac{4^3}{3!} + \frac{(-2)^3}{3!}$$

$$\vdots$$

$$x^n = \frac{4^n}{n!} + \frac{(-2)^n}{n!}$$

Q132. Solution

Correct Answer: (A)

Here, $v_1 = \frac{v}{\left(\frac{40}{195}\right)}$ and $v_2 = \frac{v}{\left(\frac{40}{193}\right)}$ $\Rightarrow$ $v_1 = \frac{v(195)}{40}$ and $v_2 = \frac{v(193)}{40}$ According to given condition

$$\frac{v}{40}(195) - v = 9 \qquad \ldots(i)$$

and

$$v - \frac{v\,(193)}{40} = 9 \qquad \ldots(ii)$$

Adding Eqs. (i) and (ii) $18 = \frac{v}{40}(195 - 193) = \frac{2v}{40} \Rightarrow 18 = \frac{v}{20}$

or $v = 360 \text{ m/s}$

Q133. Solution

Correct Answer: (A)

Reaction of chloroform and ethanolic potassium hydroxide with amine to give isocyanides is known as carbylamine reaction. The necessary condition is that amine should be a primary amine.

Q134. Solution

Correct Answer: (C)

Refractive index of material of prism $\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}}$ but $\delta_m = A$ $\mu = \frac{\sin\frac{A+A}{2}}{\sin\frac{A}{2}}$ $\mu = \frac{\sin A}{\sin A/2}$

$\mu = \frac{2\sin A/2\cos A/2}{\sin A/2}$ $\cos\frac{A}{2} = \frac{\mu}{2}$ $\frac{A}{2} = \cos^{-1}\left(\frac{\mu}{2}\right)$ $A = 2\cos^{-1}\left(\frac{\mu}{2}\right)$

Q135. Solution

Correct Answer: (B)

It is given that, $f(x+y) = f(x) + f(y)$ and $\begin{matrix} f(1) = 7 \\ f(x) = 7x \end{matrix}$ $\therefore$ $f(x) = 7x$ So,

$$\sum_{t=1}^{39} f(t) = 7[1 + 2 + 3 + \ldots + 39]$$

$$= 7 \times \frac{39 \times 40}{2} = 5460.$$

Q136. Solution
Correct Answer: (D)

The given statement is:

Either the orangutan is not angry, or he frowns upon the world.

According to it the ordered pair of statements are:

C. The orangutan does not frown upon the world.

B. The orangutan is not angry.

Or the statement can be:

D. The orangutan is angry.

A. The orangutan frowns upon the world.

So, the required answer is CB and DA.

Hence, this is the correct answer.

Q137. Solution
Correct Answer: (A)

The correct option is A I, II and III are correct
I. $2\mathrm{CaOCl}_2 + \mathrm{CH}_3\mathrm{COCH}_3 \rightarrow 2\mathrm{CHCl}_3 + \mathrm{Ca(OH)}_2 + \mathrm{CH}_3\mathrm{COOCa}$ II. Bleaching powder when catches fire decomposes without giving oxygen which further intensifies the fire. It was found that high grade calcium hypochlorite can be decomposed by giving out calcium chloride and oxygen at a particular temperature. III. Flourine is prepared by the electrolysis of fused acid potassium flouride at 373 K in a copper vessel using graphite electrodes. Option A is correct.

Q138. Solution
Correct Answer: (B)

Both (A) and (R) are true statement but (R) is not the correct explanation of (A). pH of a buffer solution is given by Henderson-Haselbalch equation.

$$\mathrm{pH} = \mathrm{p}K_a + \log \frac{[\text{ salt }]}{[\text{acid }]}$$
$$\therefore \quad \mathrm{pH} = \mathrm{p}K_a(\text{ when }[\text{ salt }] = [\text{ acid }])$$

Q139. Solution

Correct Answer: (B)

$2Z5,\ 7Y7,\ 14X9,\ 23W11,\ 34V13,$ _____

Here, the pattern is like,

$2+5=7$	Y	$5+2=7$
$7+7=14$	X	$7+2=9$
$14+9=23$	W	$9+2=11$
$23+11=34$	V	$11+2=13$
$34+13=47$	U	$13+2=15$

Using the same pattern, the next place will be filled by,

$47U15$.

Q140. Solution

Correct Answer: (A)

As we know 29th February falls in a leap year and leap year is the year that is a multiple of 4 eg. 2004, 2008, 2012 and so on.

So 29th February or leap year falls in a century 25 times.

As we know tuesday repeat after 7 days in a week.

So In a leap year tuesday repeat after 28 year.

So in a century 29th February falls in 3 times.

Q141. Solution

Correct Answer: (B)

Let the equation of plane passing through $(1,1,1)$ is $a(x-1)+b(y-1)+c(z-1)=0\ldots$ (i) It is also passing through $(1,-1,-1)$. $\therefore\quad a(0)+b(-2)+c(-2)=0$ or $b+c=0\ldots$ (ii) Since, Eq. (i) is perpendicular to the plane $2x-y+z+5=0$ $\therefore\quad 2a-b+c=0\ldots$ (iii) Since, Eqs. (ii) and (iii) are identical $\therefore\quad \frac{a}{-2-2}=\frac{b}{-4-0}=\frac{c}{4}\Rightarrow\frac{a}{-4}=\frac{b}{-4}=\frac{c}{4}$ $\therefore$ Required equation of plane is $x+y-z-1=0$

Q142. Solution

Correct Answer: (B)

$$\text{Given, } A(\alpha,\beta)=\begin{bmatrix}\cos\alpha & \sin\alpha & 0\\ -\sin\alpha & \cos\alpha & 0\\ 0 & 0 & e^{\beta}\end{bmatrix}$$

$$\text{Now, } |A(\alpha,\beta)|=e^{\beta}\left(\cos^2\alpha+\sin^2\alpha\right)=e^{\beta}$$

$$C_{11}=e^{\beta}\cos\alpha, C_{12}=e^{\beta}\sin\alpha, C_{13}=0$$

$$C_{21}=-e^{\beta}\sin\alpha, C_{22}=e^{\beta}\cos\alpha, C_{23}=0$$

$$C_{31}=0, C_{32}=0, C_{33}=\cos^2\alpha+\sin^2\alpha=1$$

$$\therefore \operatorname{adj}(A(\alpha,\beta))=\begin{bmatrix}e^{\beta}\cos\alpha & -e^{\beta}\sin\alpha & 0\\ e^{\beta}\sin\alpha & -e^{\beta}\cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$\therefore [A(\alpha,\beta)]^{-1}=\frac{1}{e^{\beta}}\times e^{\beta}\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & e^{-\beta}\end{bmatrix}$$

$$=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & e^{-\beta}\end{bmatrix}=A(-\alpha,-\beta)$$